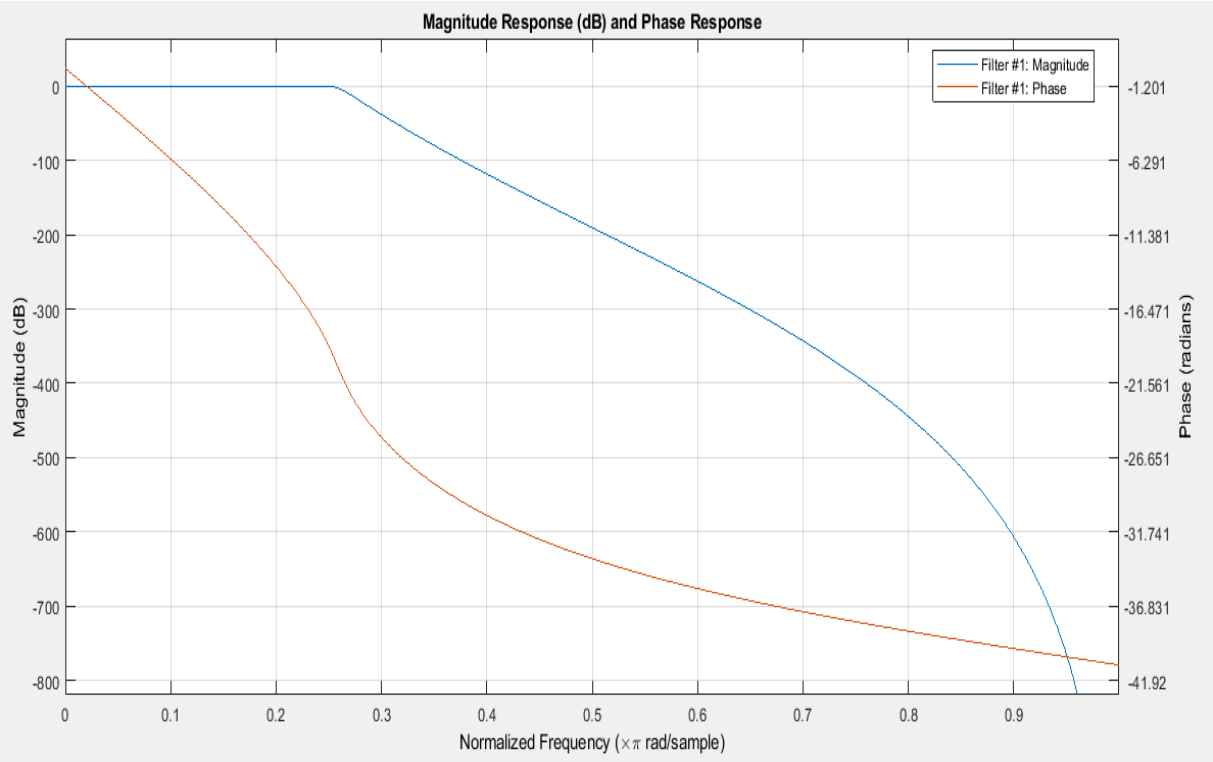
BOTSWANA INTERNATIONAL UNIVERSITY OF SCIENCE & TECHNOLOGY

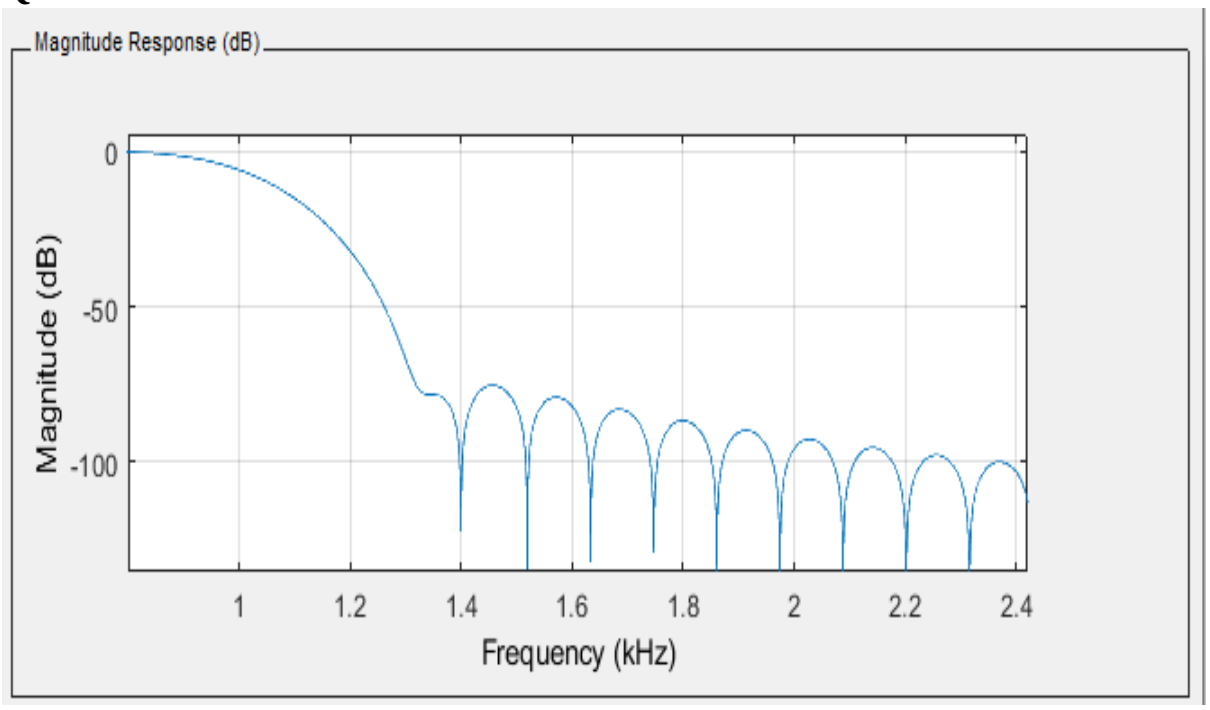
NAME	SURNAME	STUDENT ID	TIME SPENT
ISHEANESU	**MUTAMBA**	**20000721**	**24hrs**
SIMANGALISO	**MDLULI**	**21000300**	**24hrs**

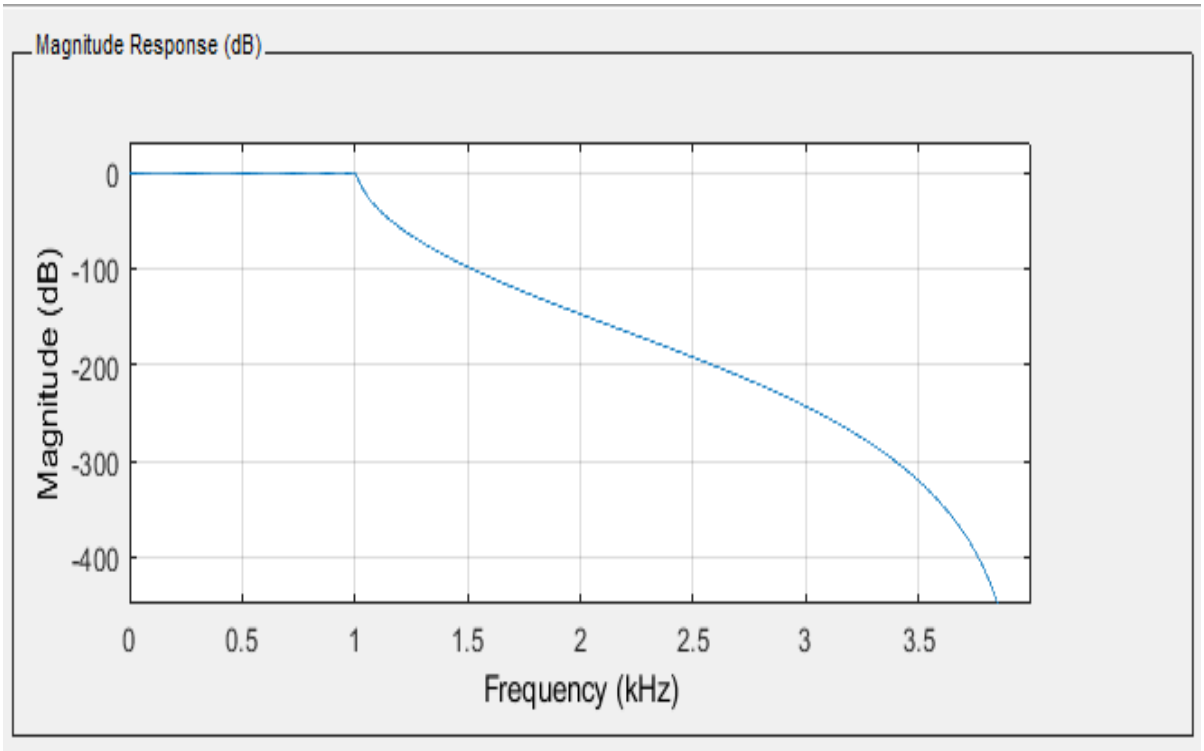
1) Butterworth filter



2) CHEBYSHEV

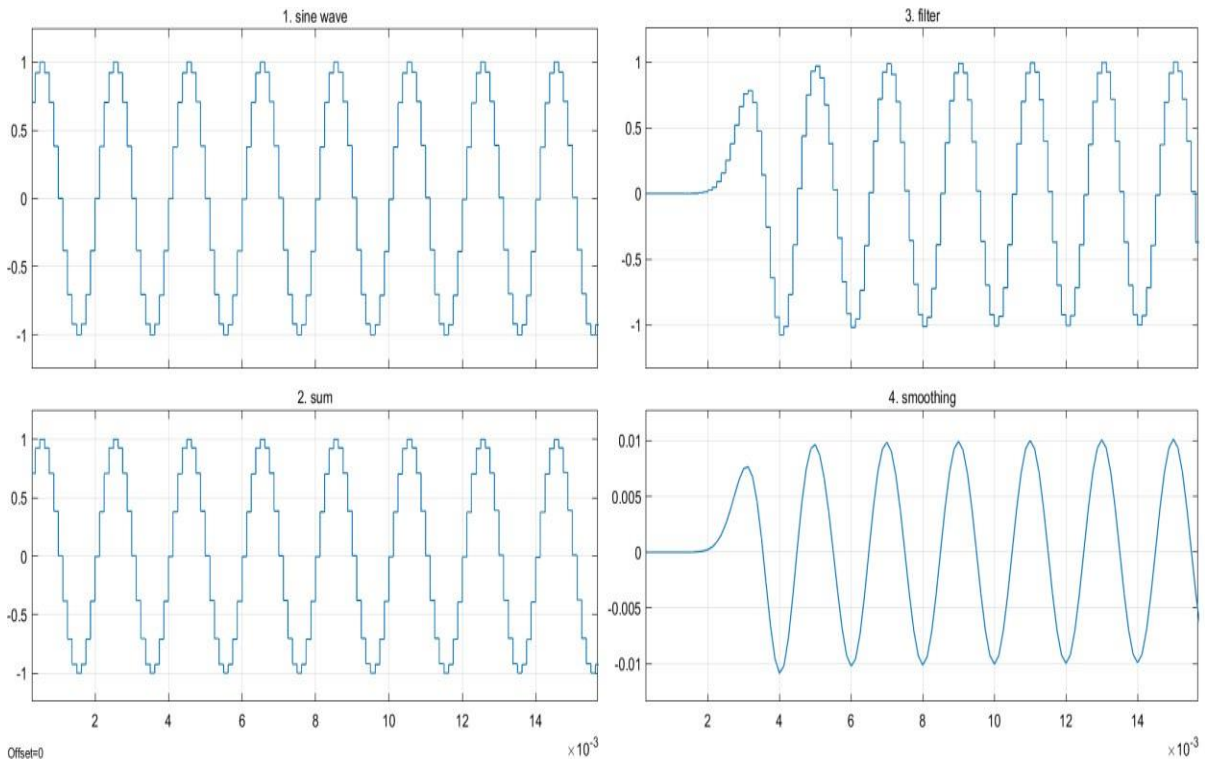
Q3 black window



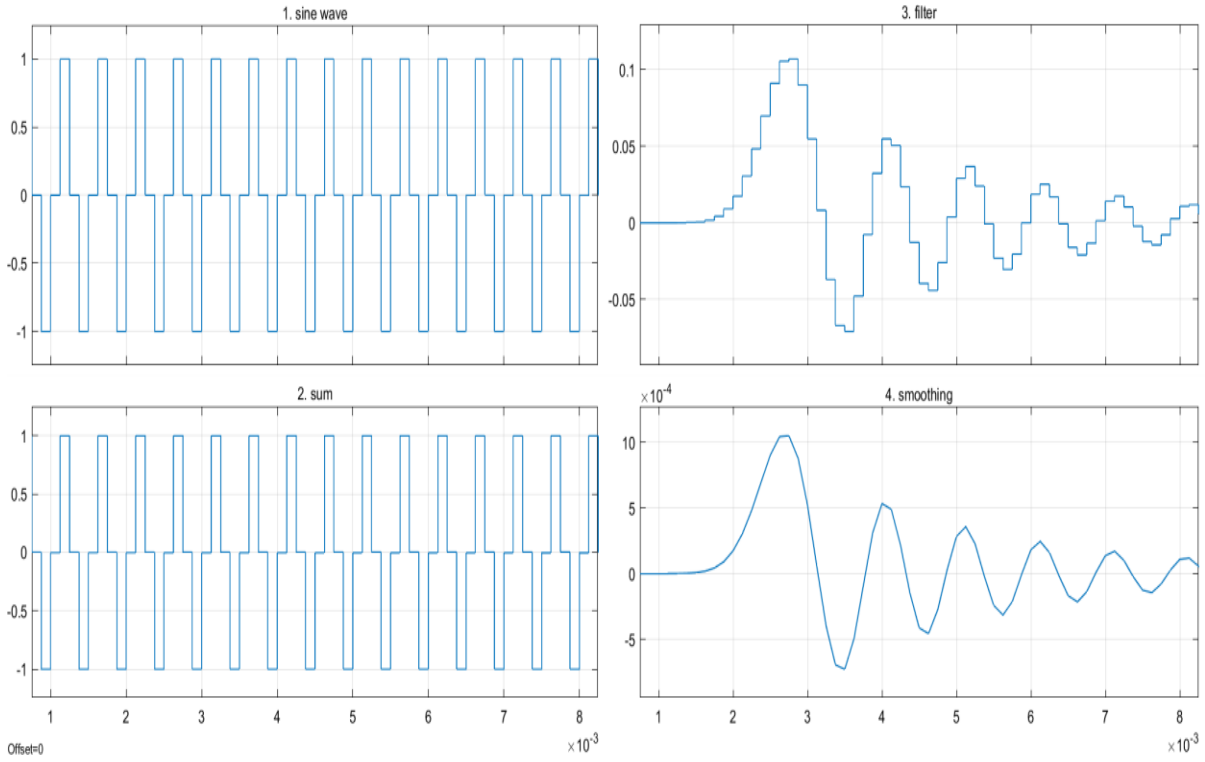


Butterworth (1)

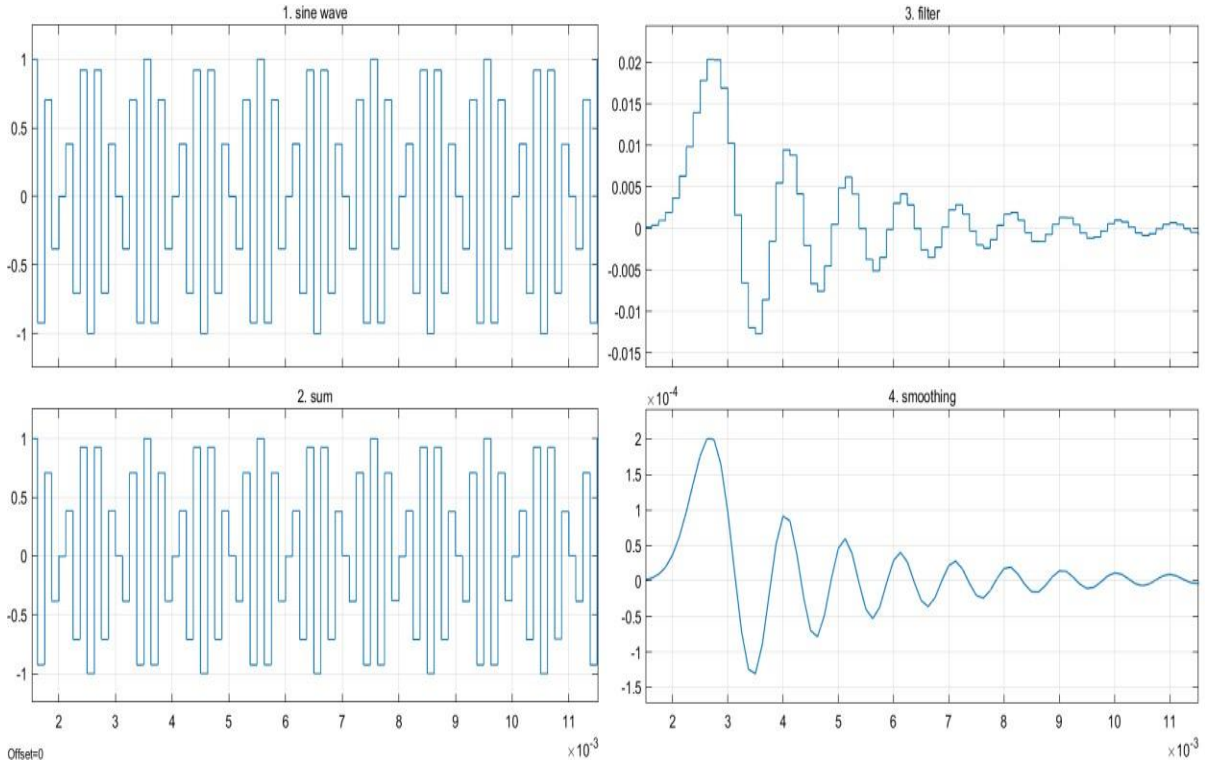
Passband 0.5kHz



Transition region 1.1 kHz

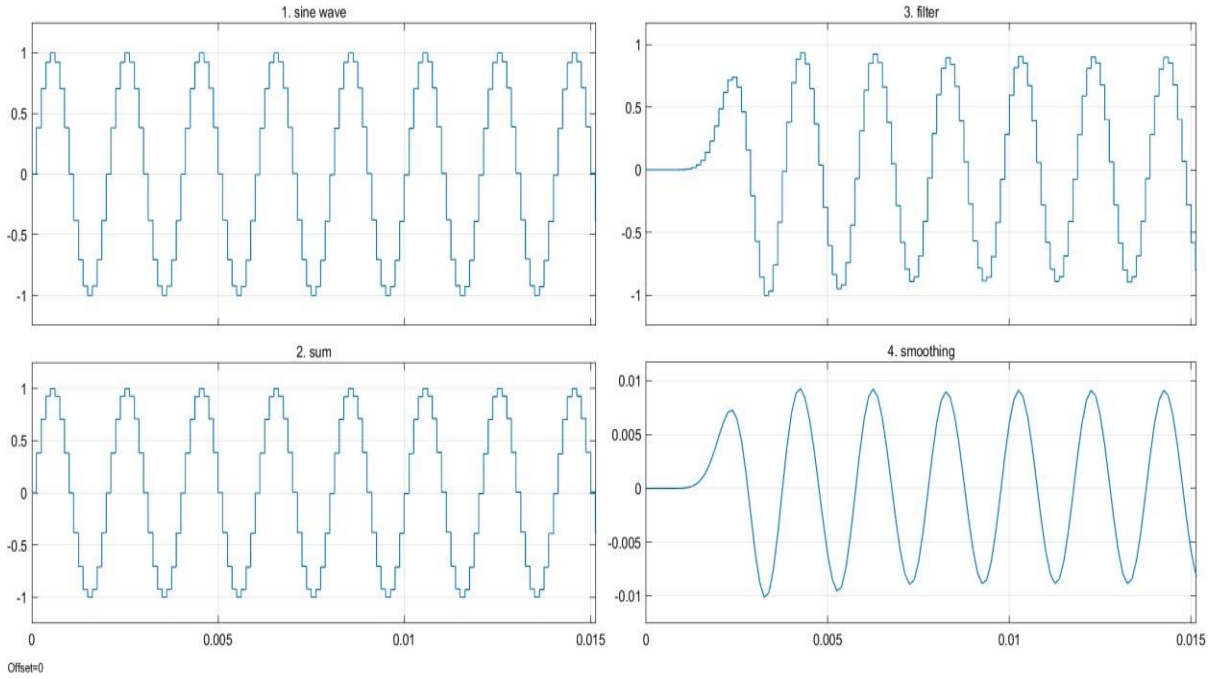


Stopband- 3kHz

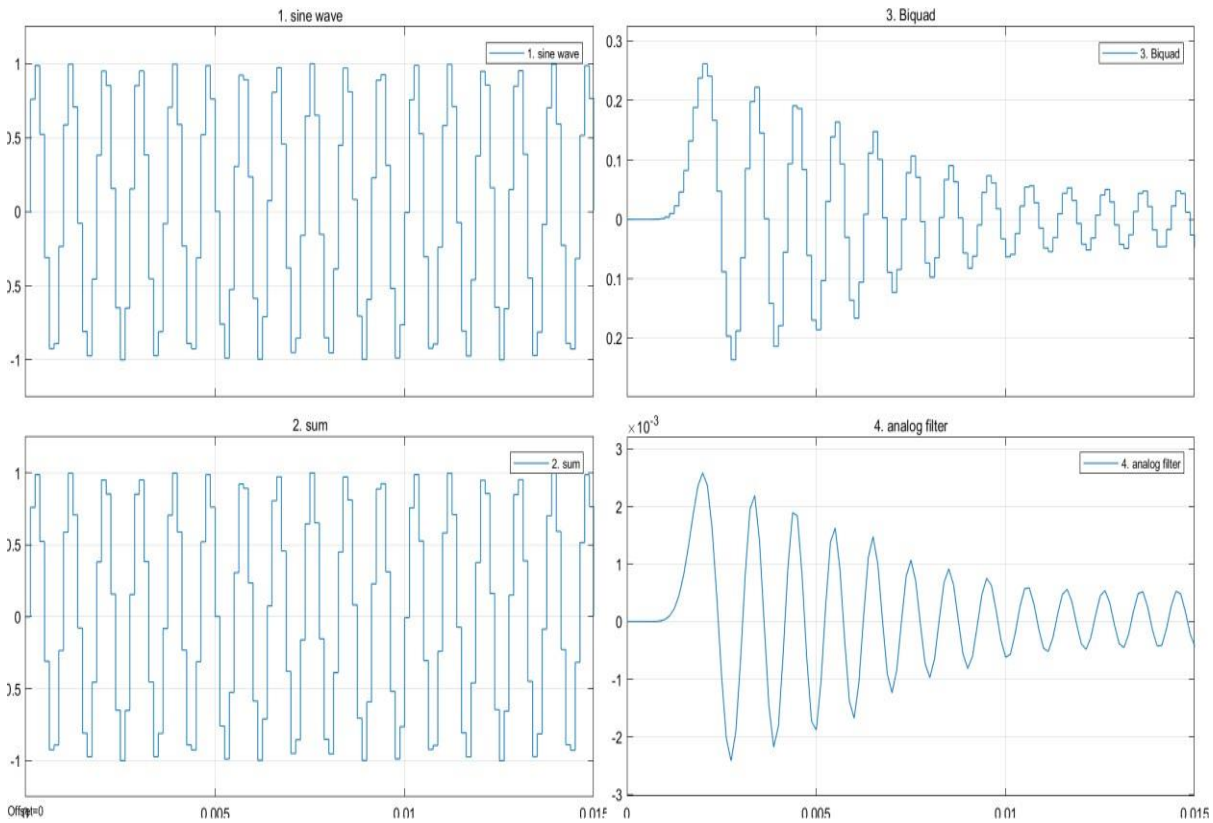


CHEBYSHEV

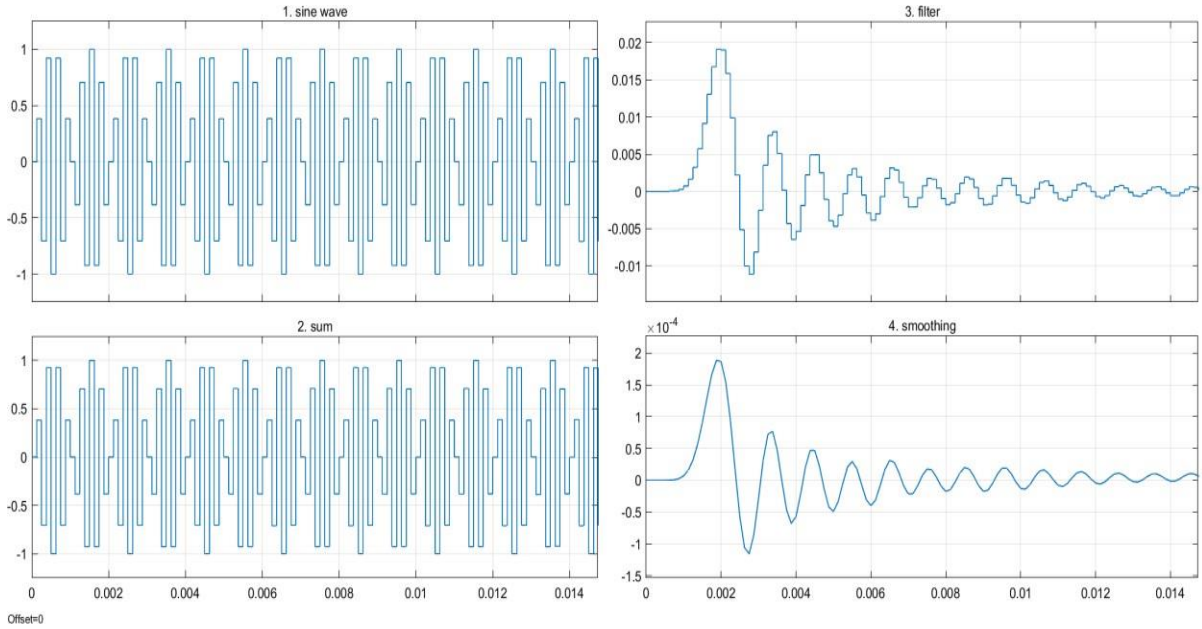
Passband – 0.5kHz



Transition – 1.5kHz

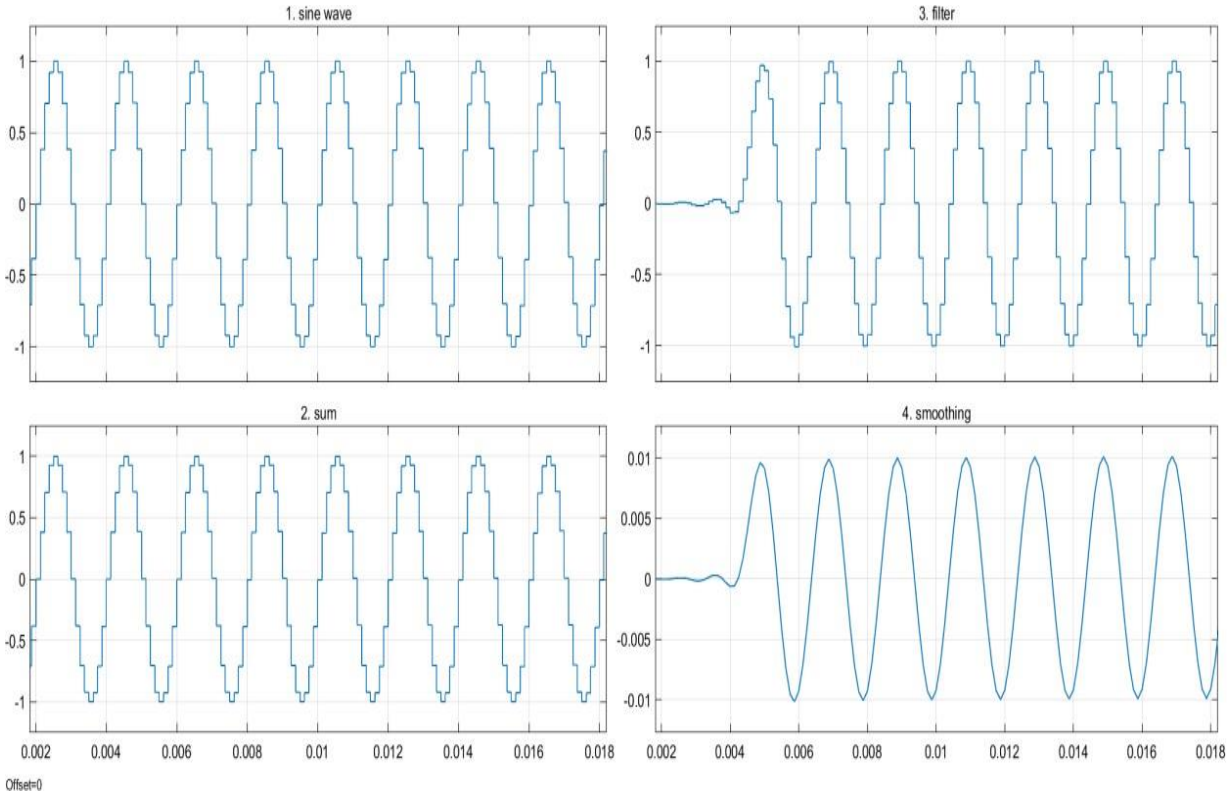


Stopband 3.5kHz

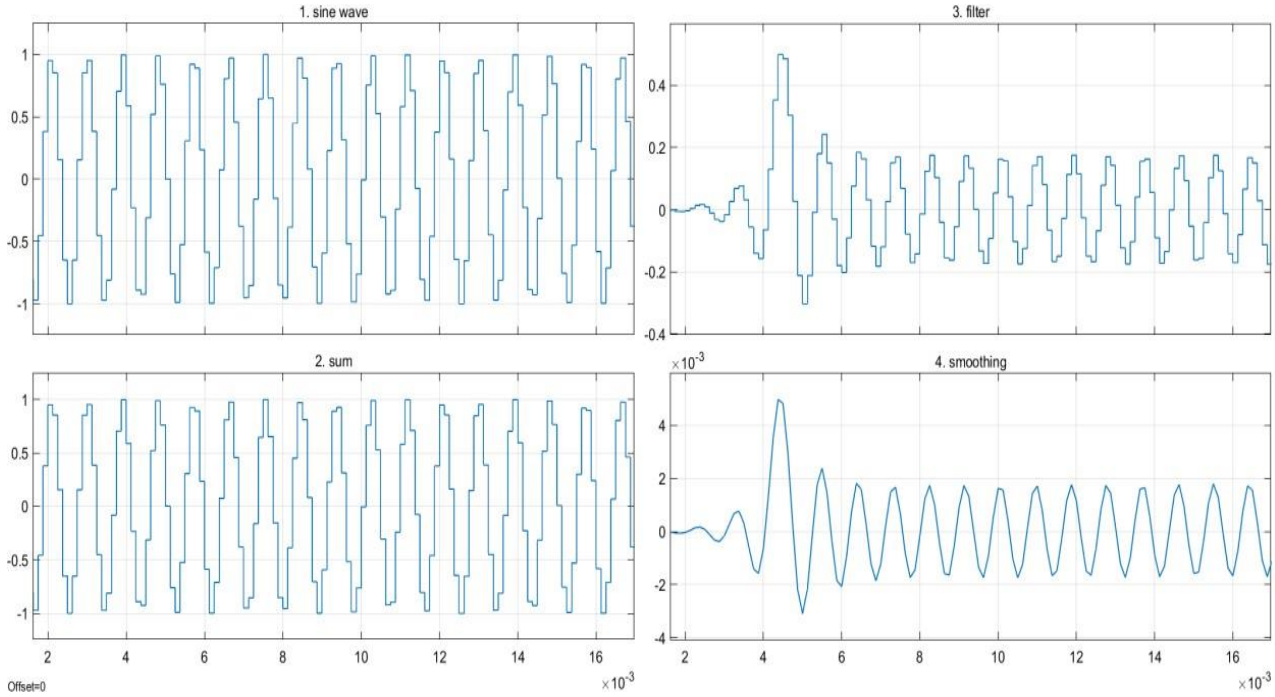


FIR

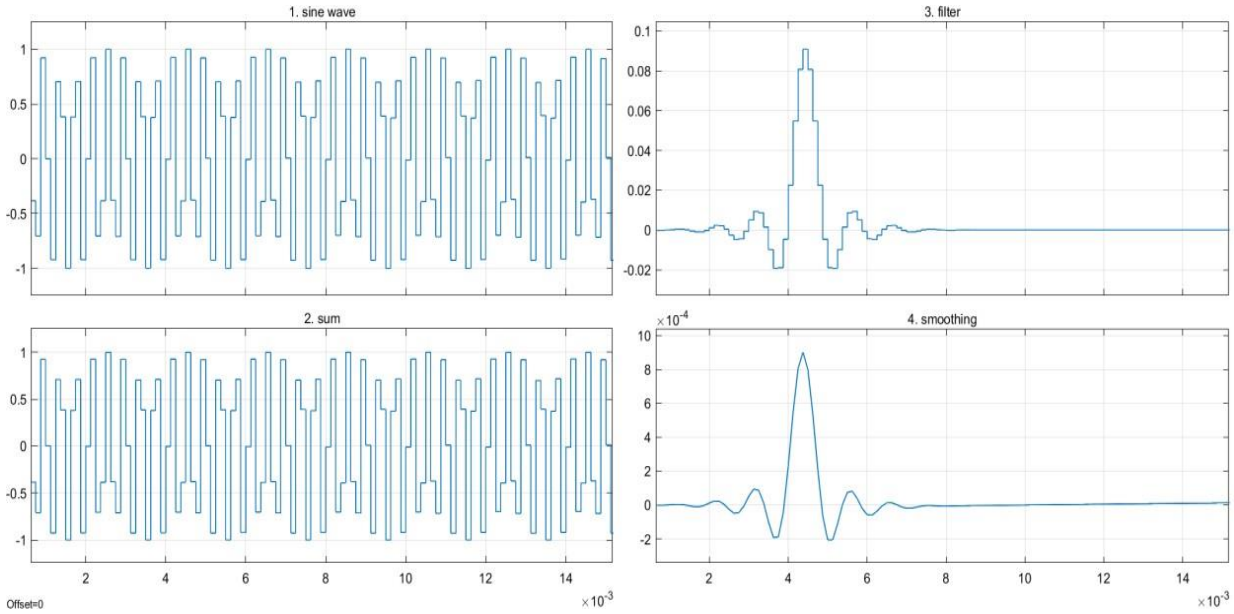
Passband – 0.5kHz



Transition – 1.1kHz



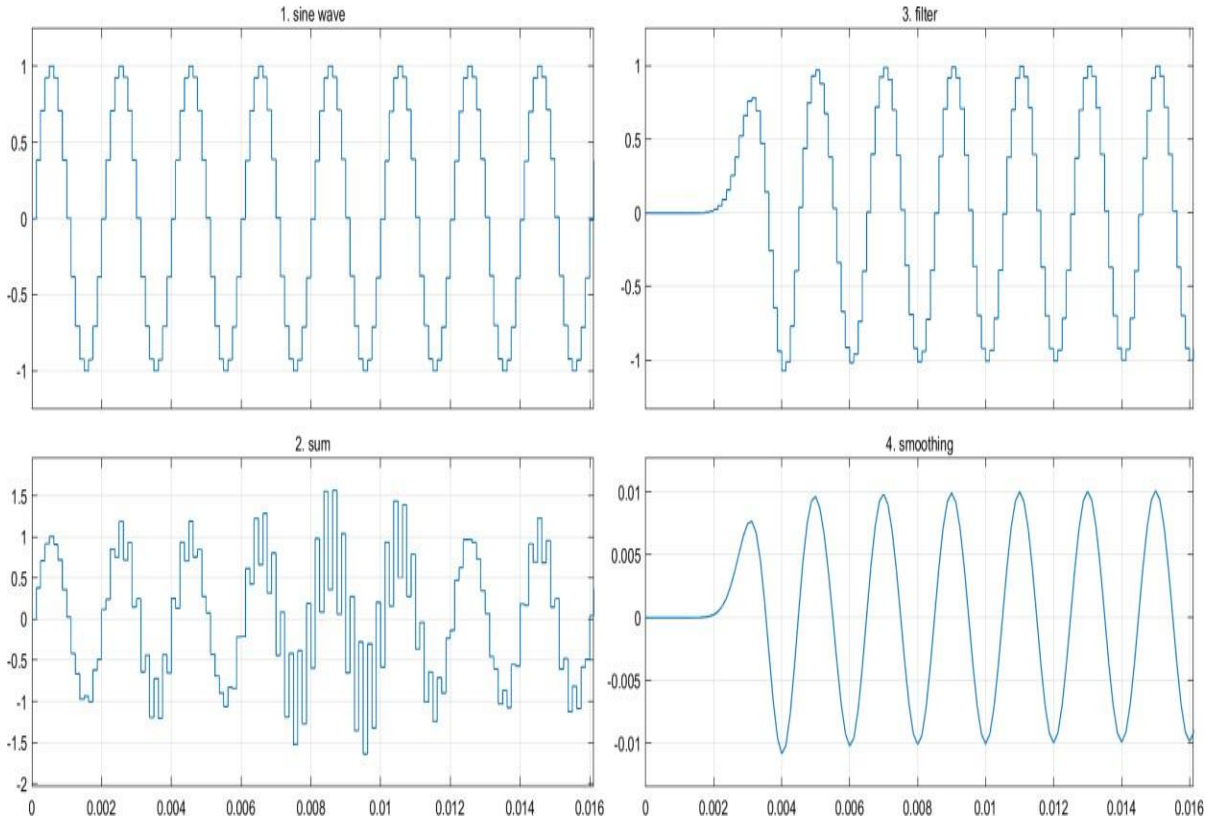
Stopband 2.5kHz



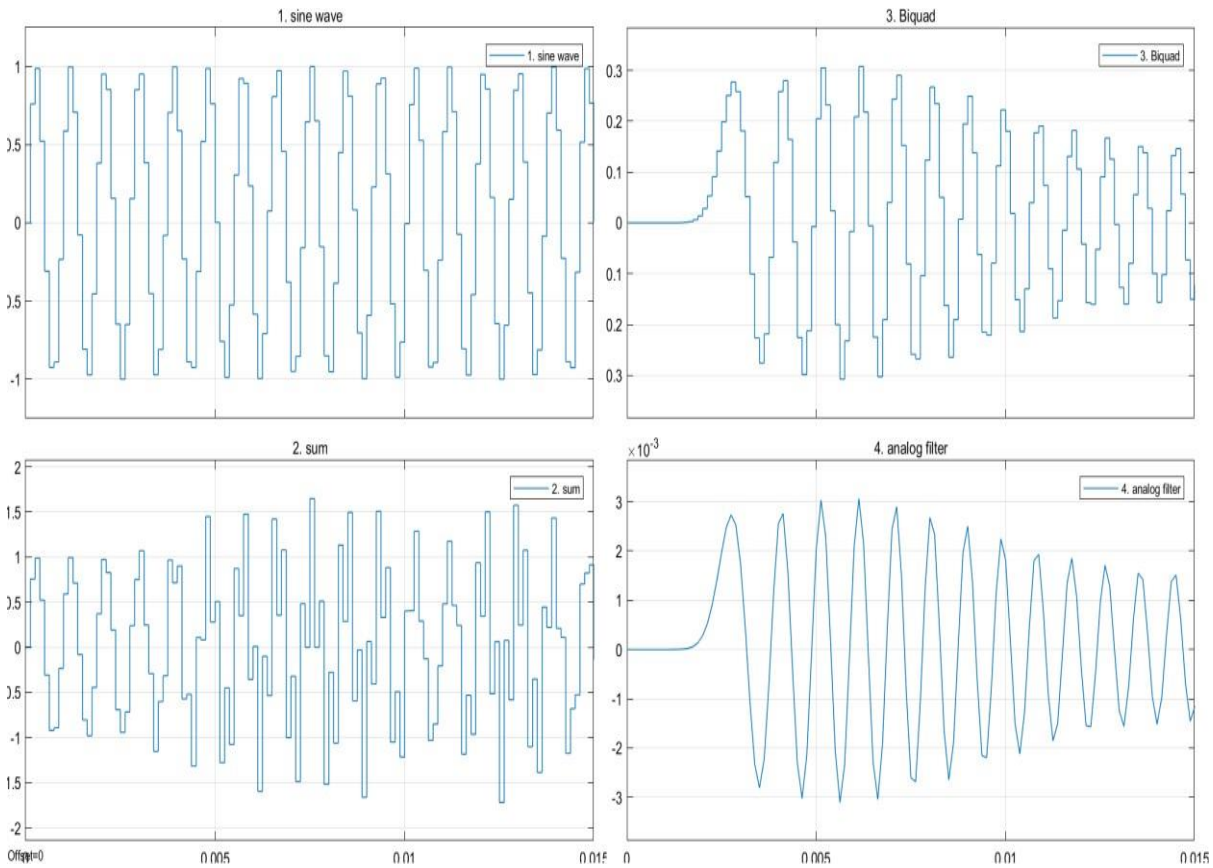
Butterworth

HIGHPASS FILTER

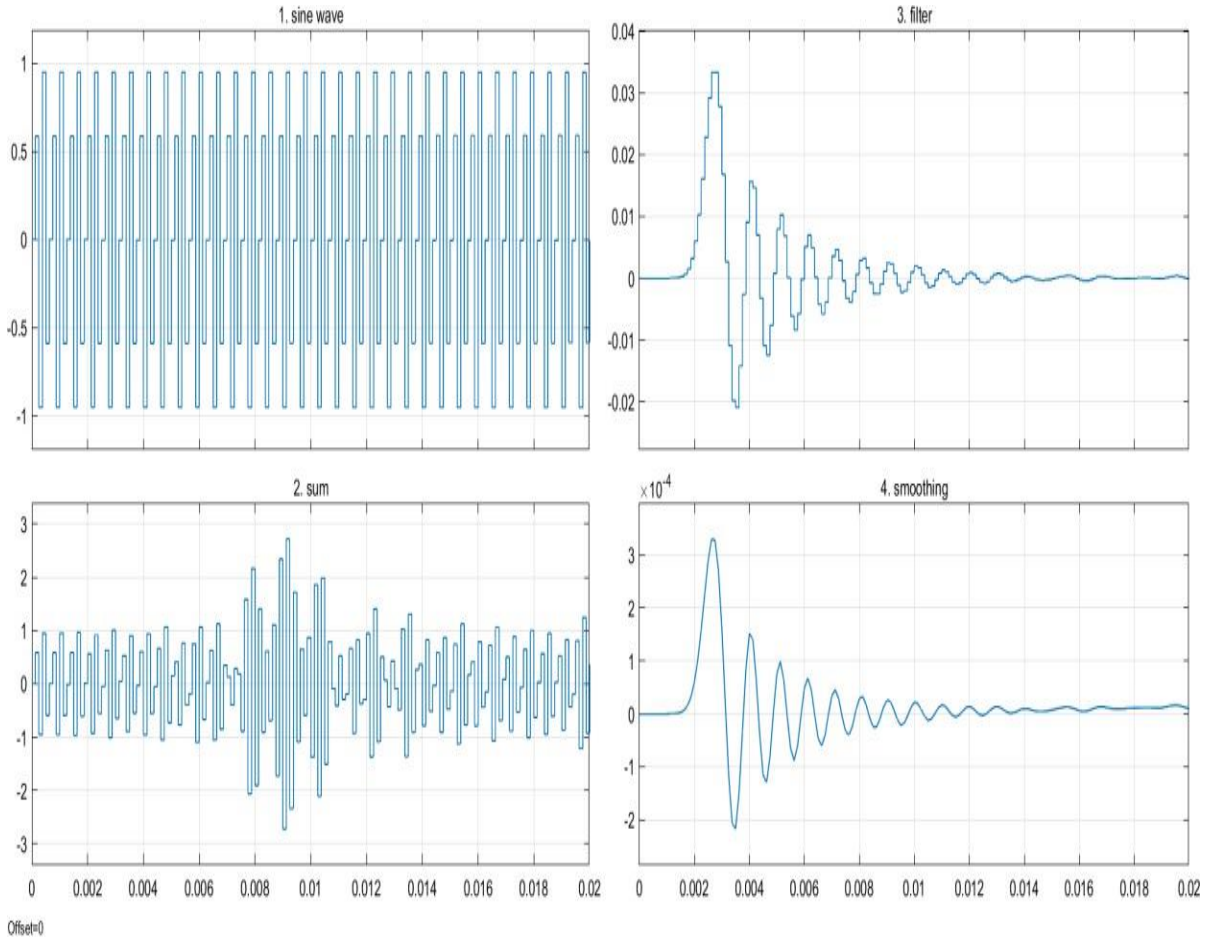
Passband – 0.5kHz



Transition – 1.1kHz

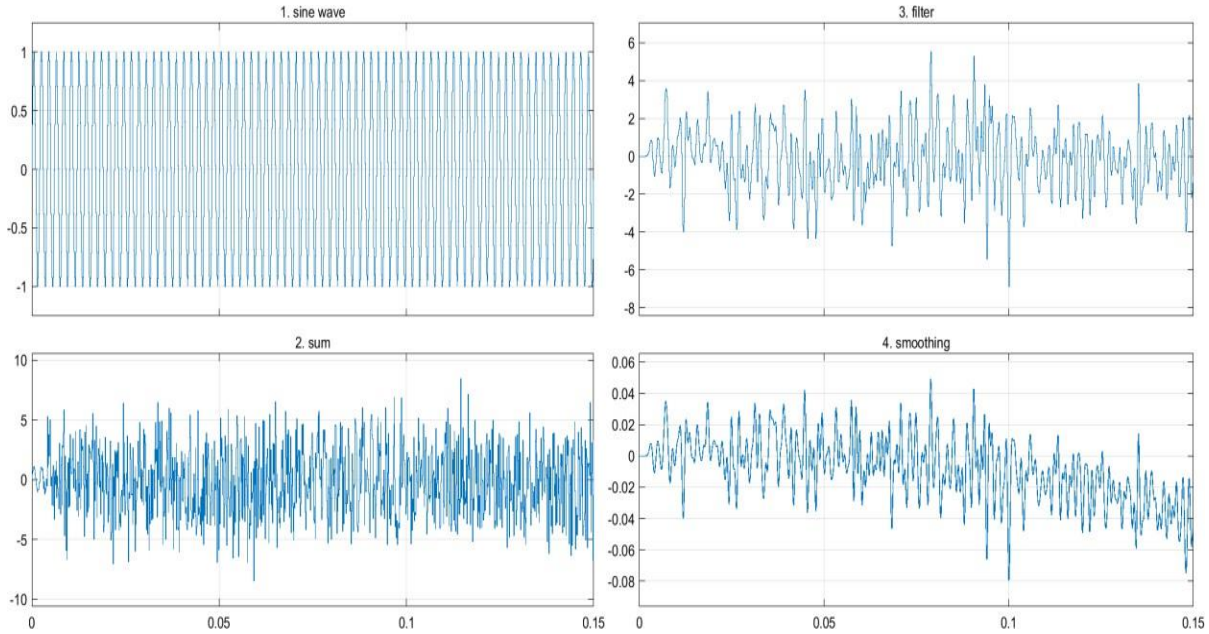


Stopband – 3.2kHz

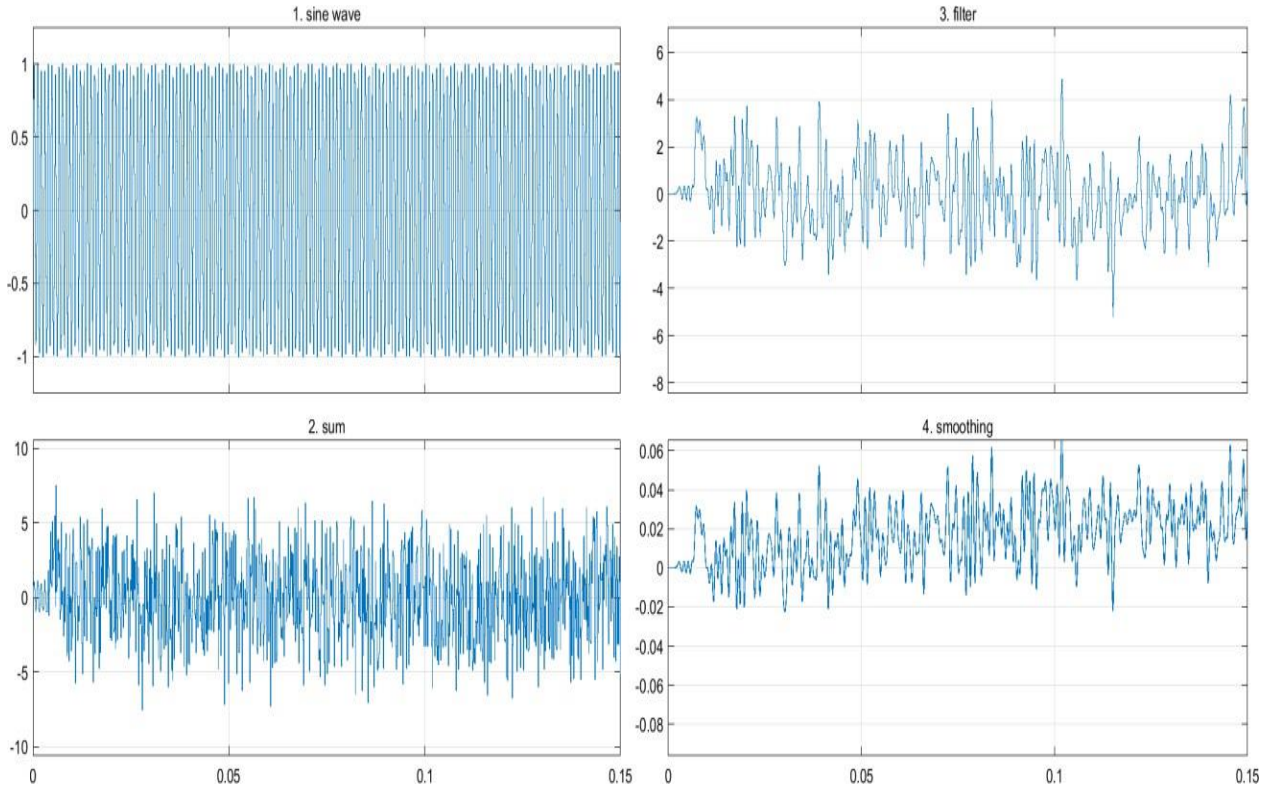


LOWPASS FILTER --BUTTERWORTH

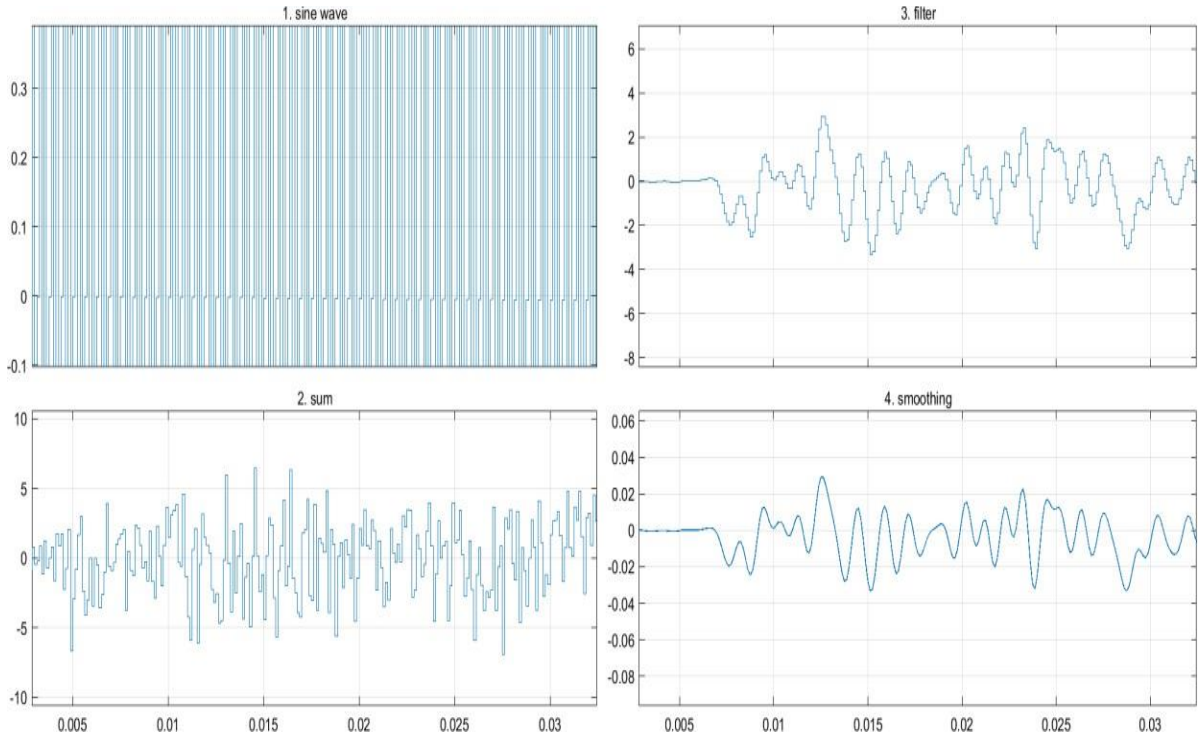
Passband -0.5kHz



Transition – 1.1kHz

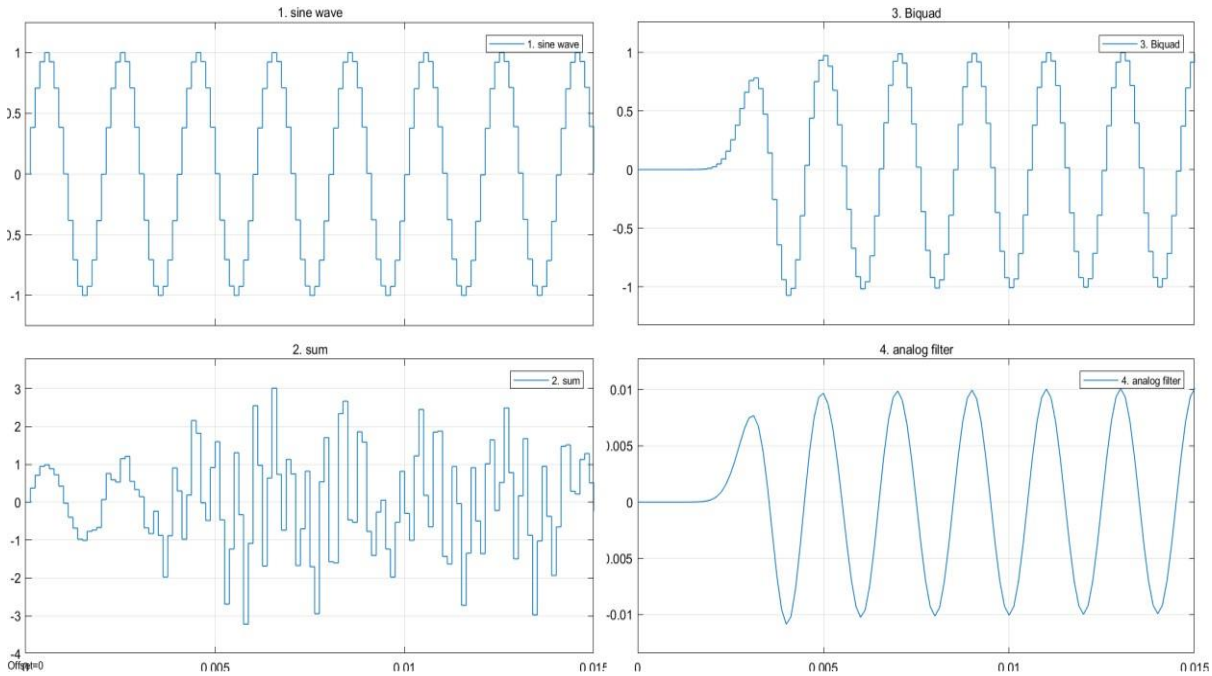


Stopband – 3.5kHz

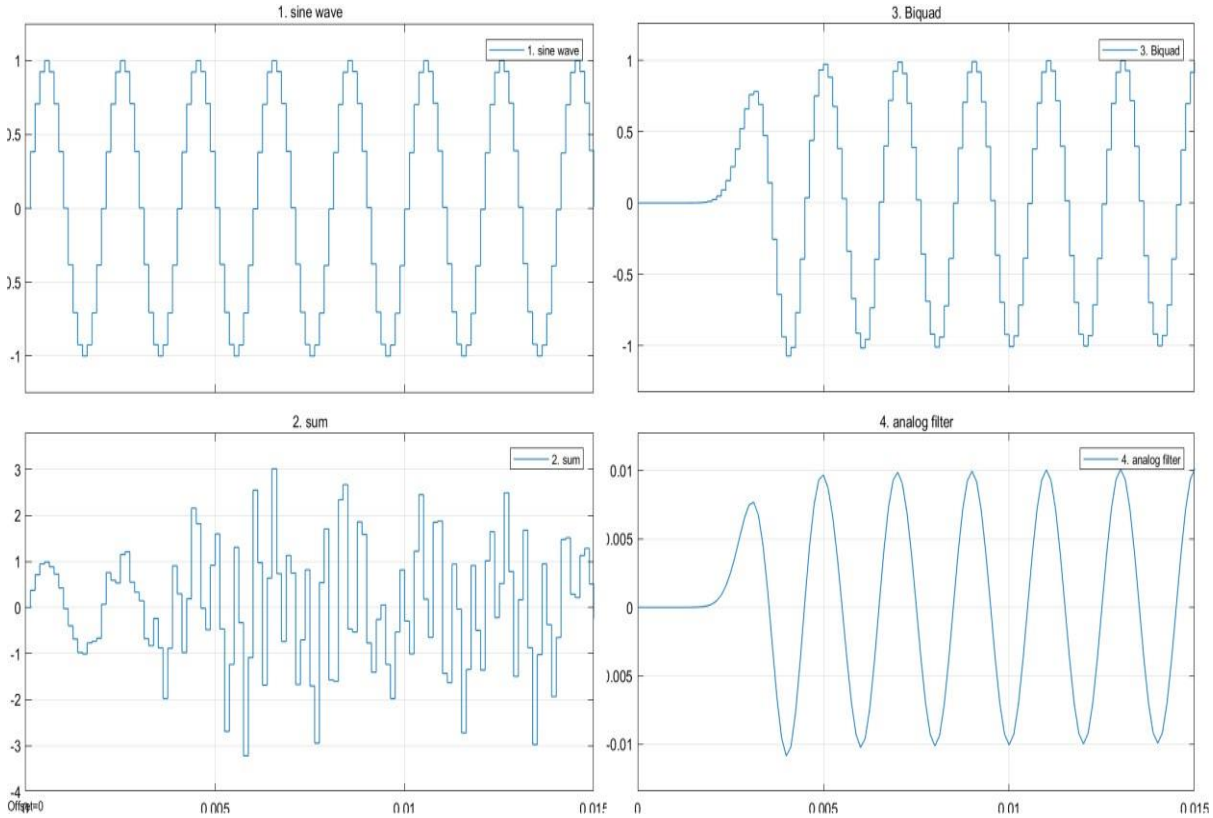


BANDPASS FILTER --BUTTERWORTH

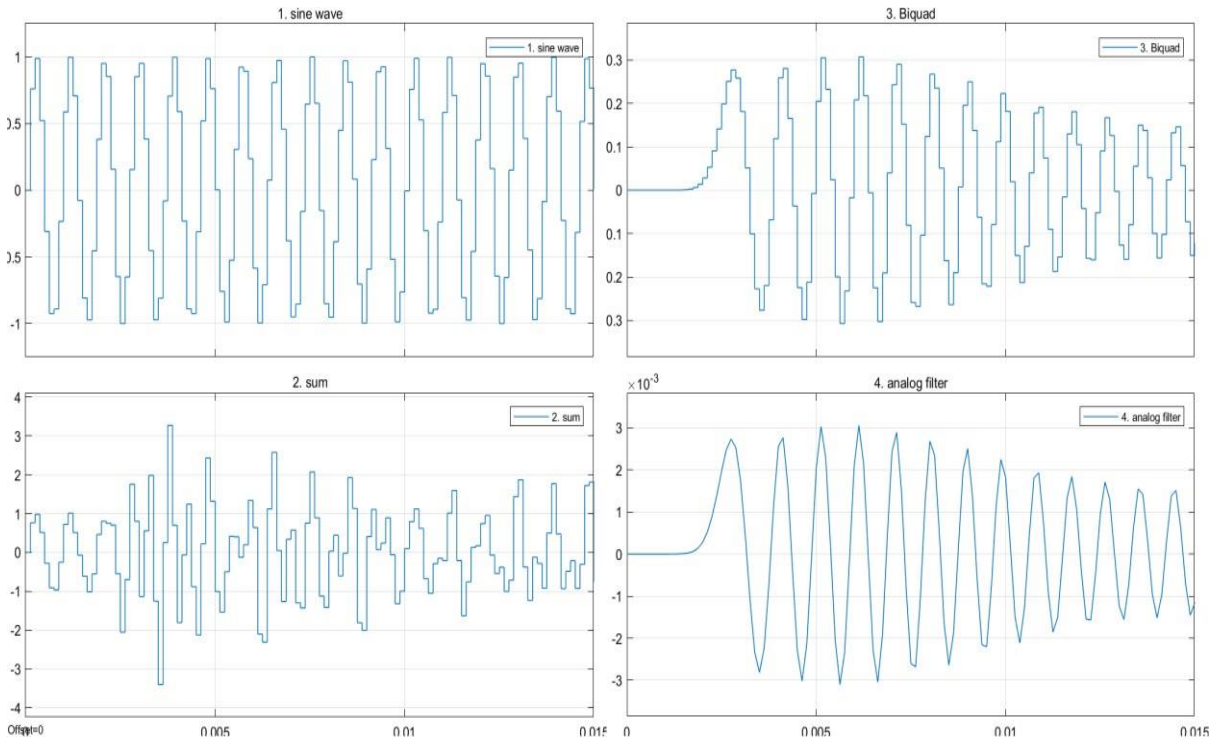
Passband – 0.5kHz



Transition – 1.1kHz



Stopband – 3.5kHz

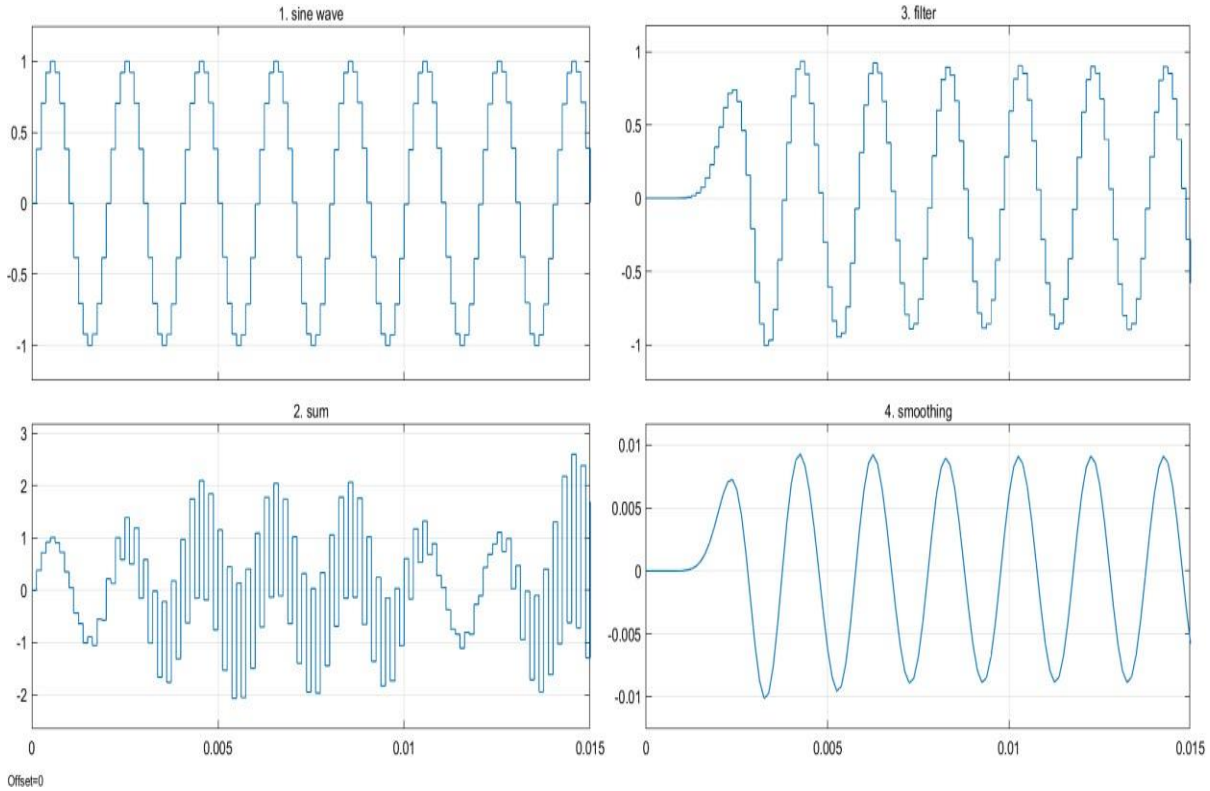


Chebyshev

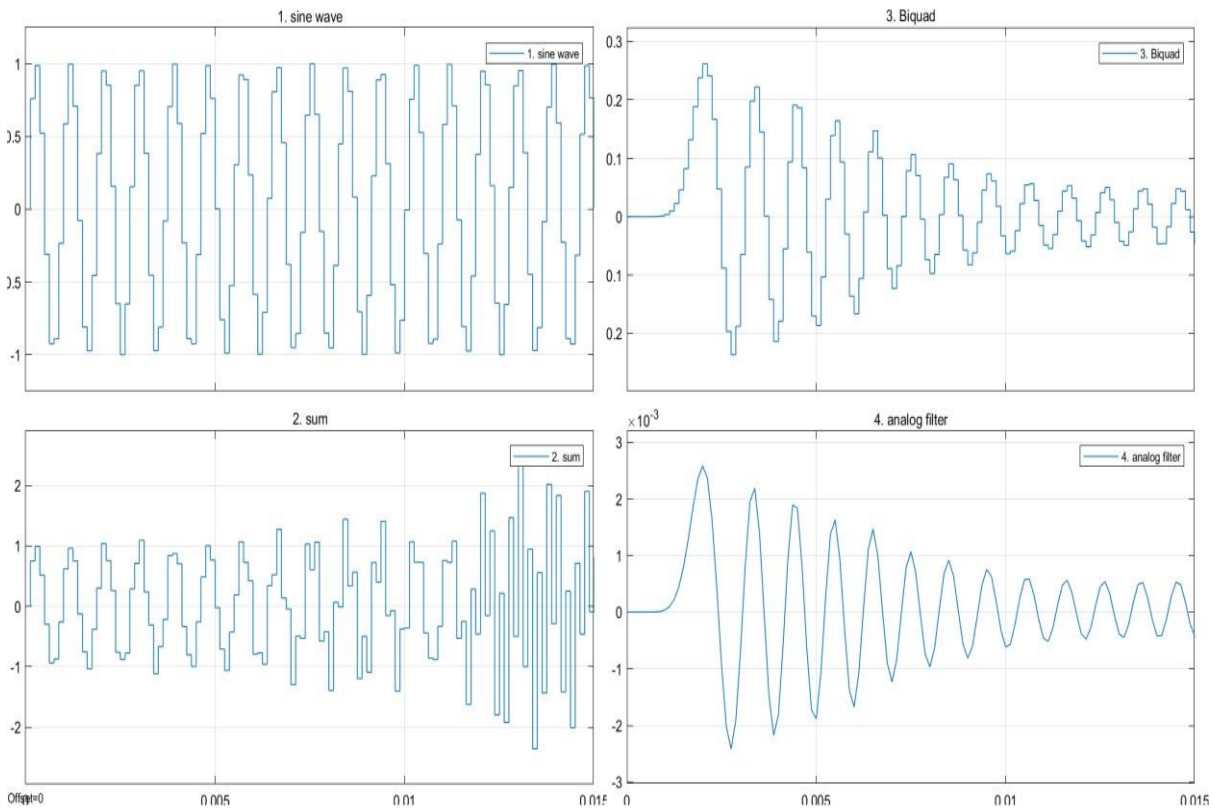
HIGHPASS

FILTER

Passband 0.5kHz



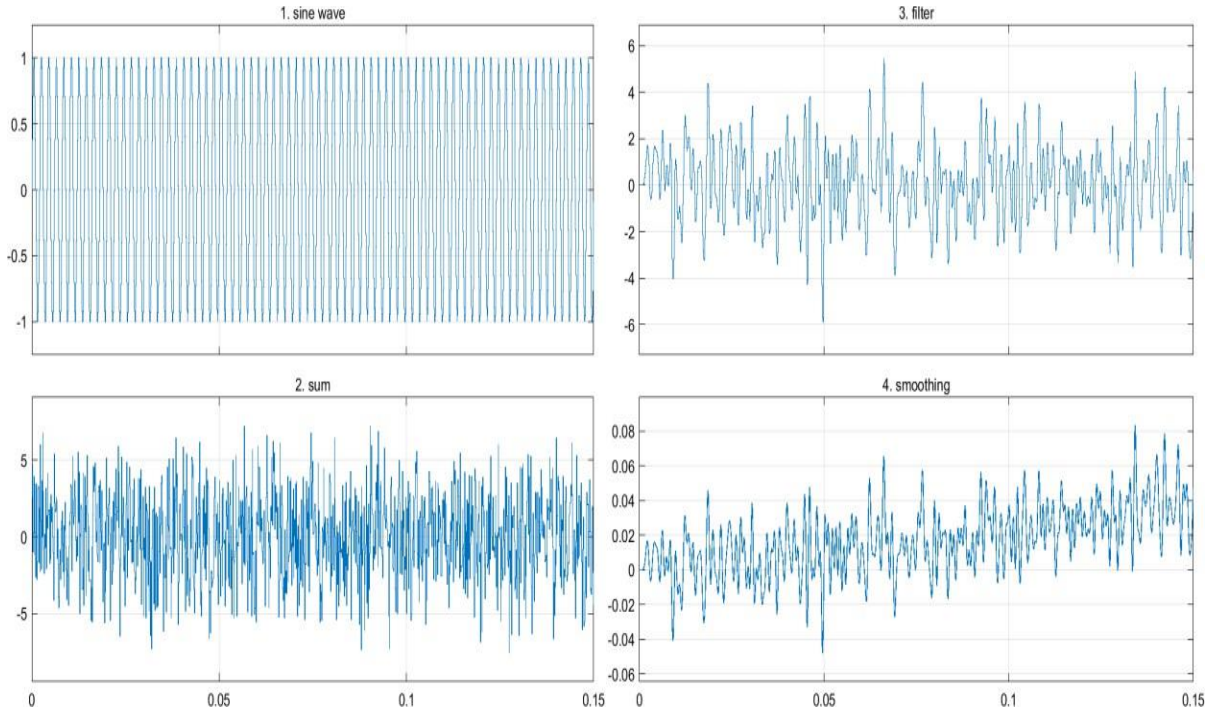
Transition – 1.1kHz



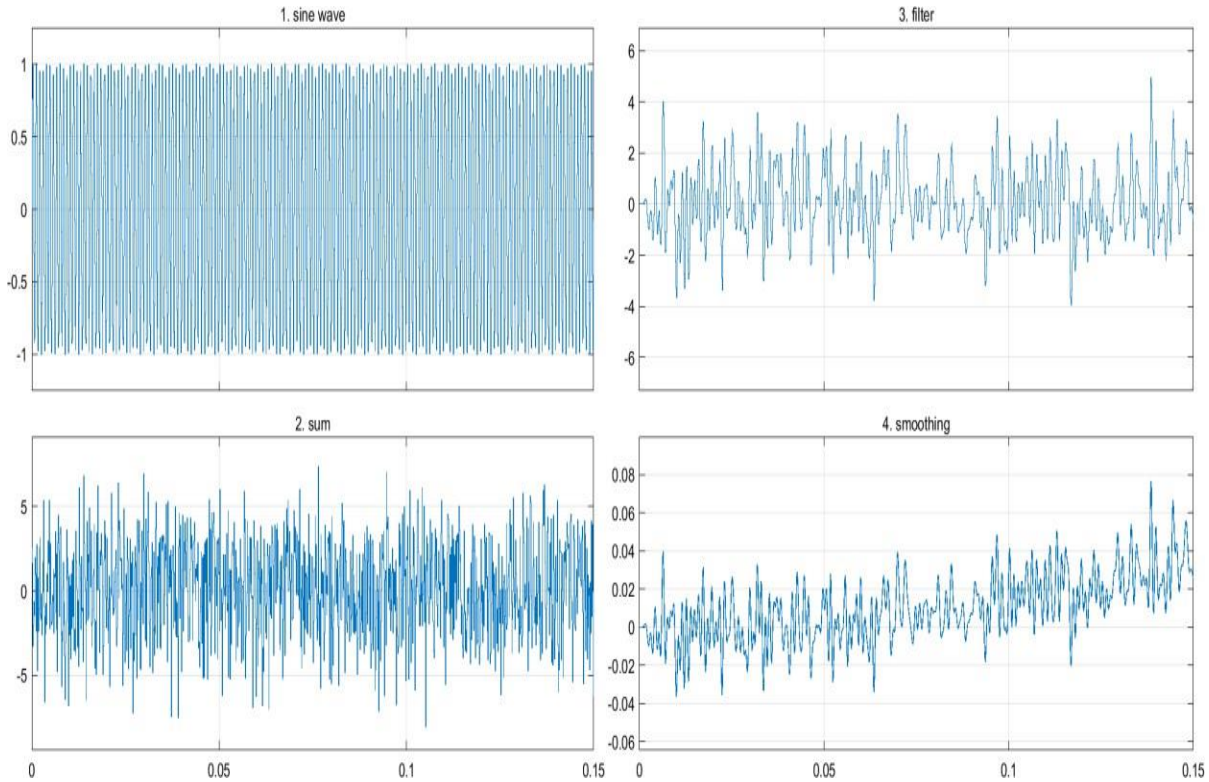
Stopband – 3.5kHz

LOWPASS FILTER --CHEBYSHEV

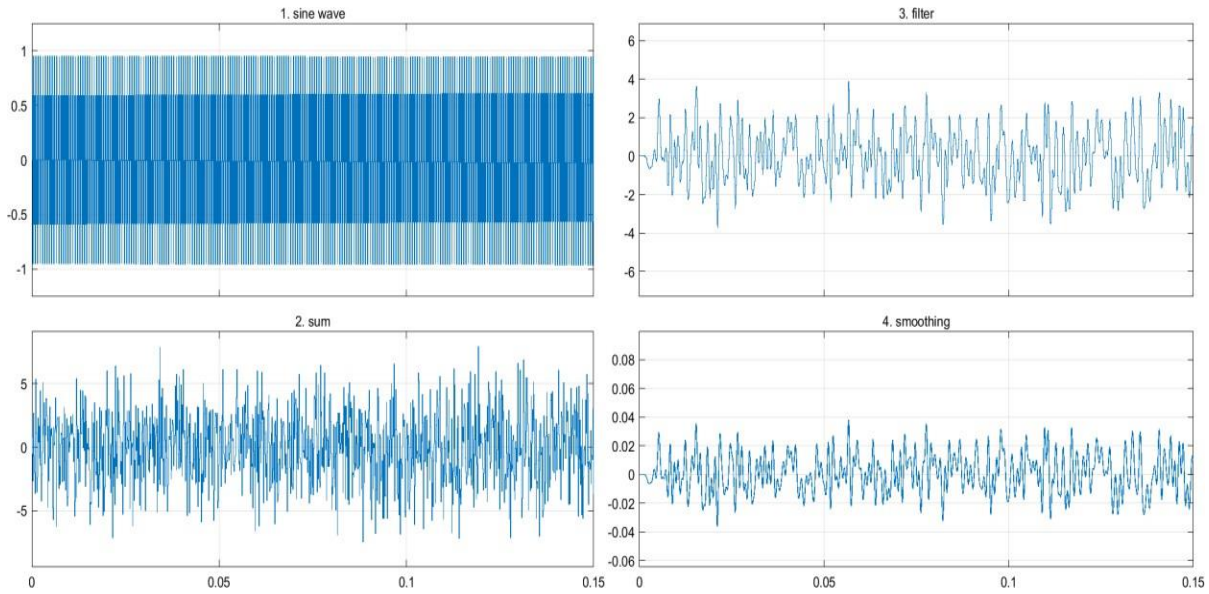
Passband



Transition

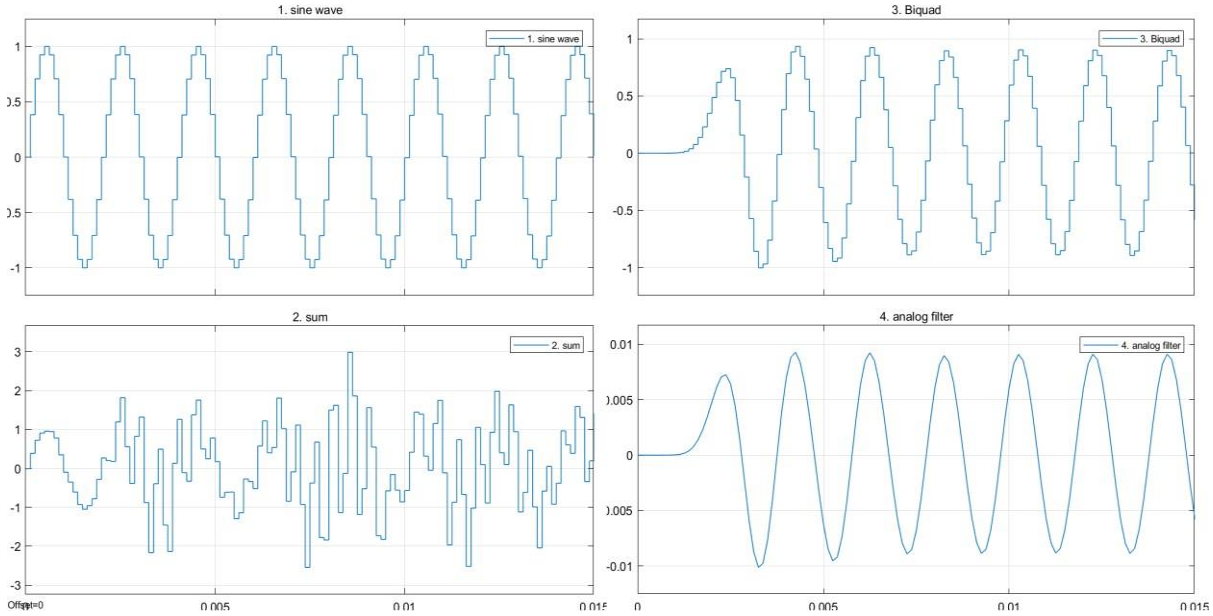


Stopband

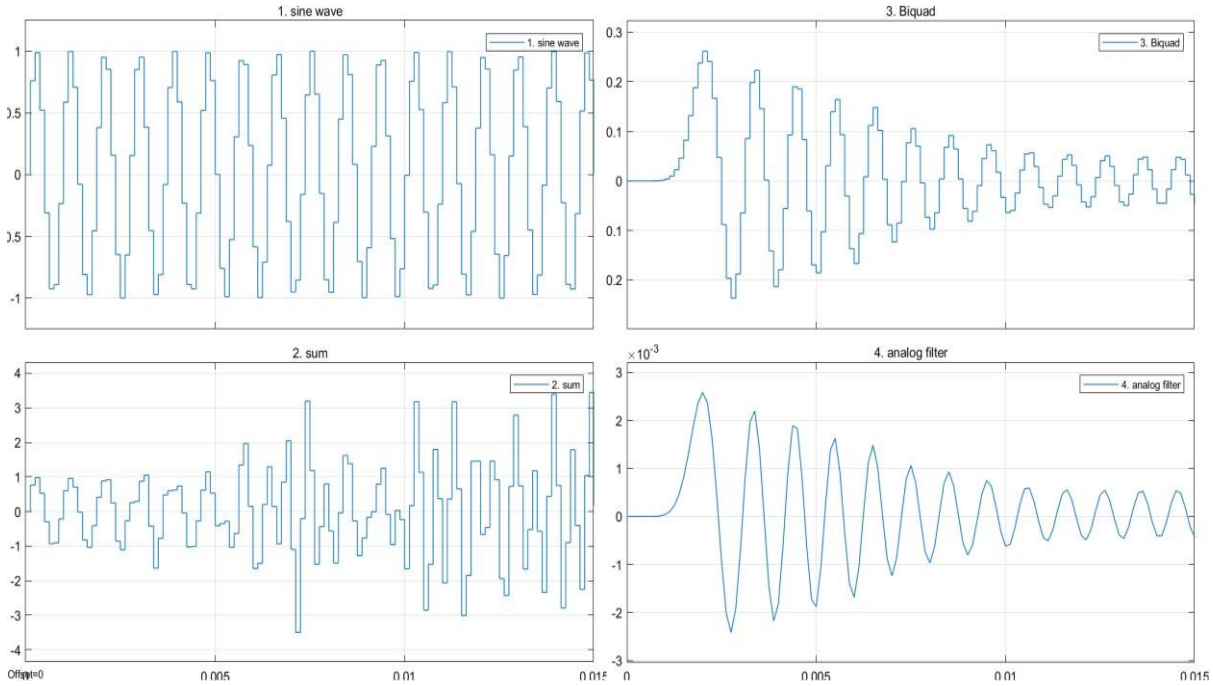


BANDPASS FILTER --CHEBYSHEV

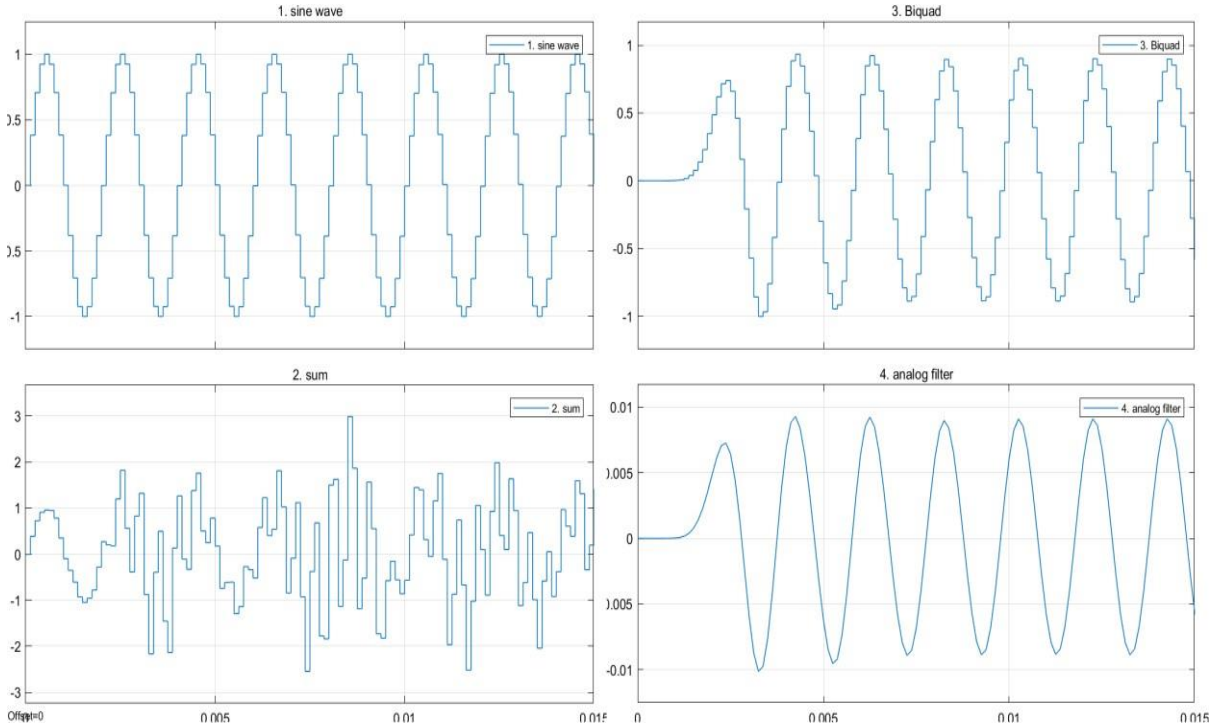
Passband – 0.5kHz



Transition – 1.1kHz

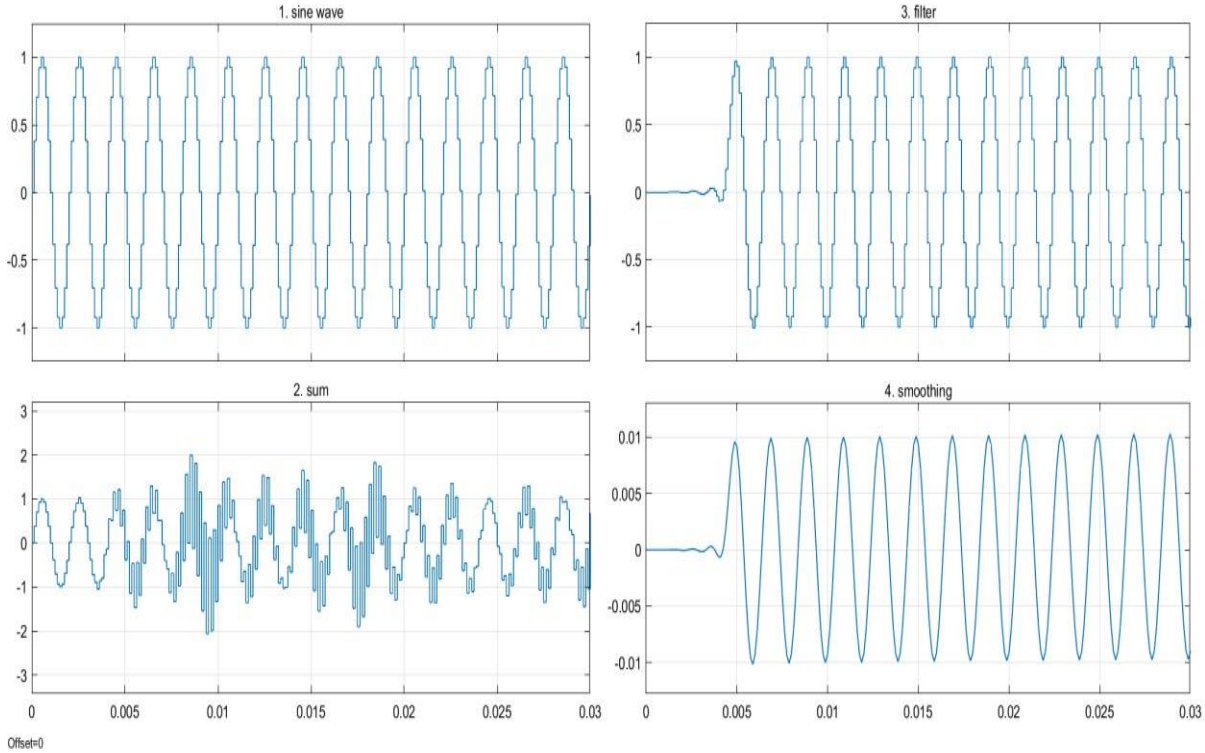


Stopband – 3.5kHz

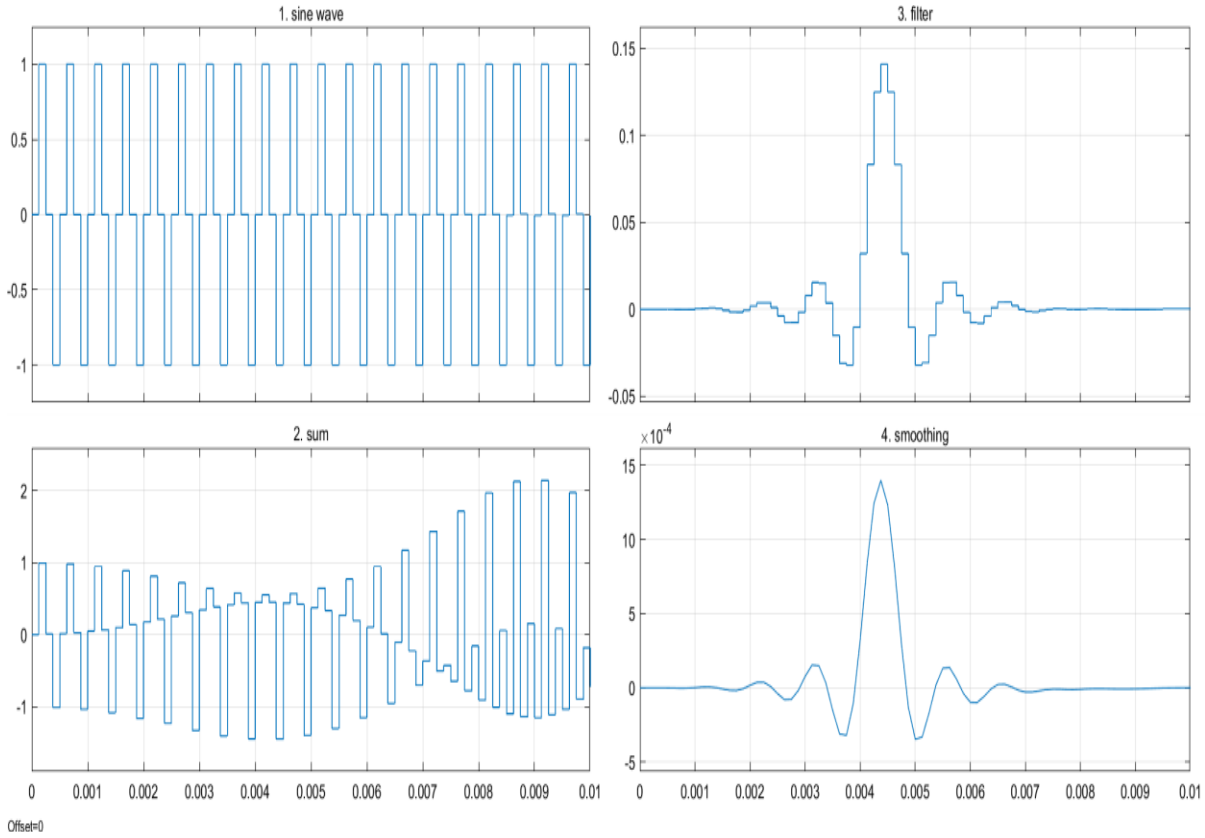


HIGHPASS FILTER

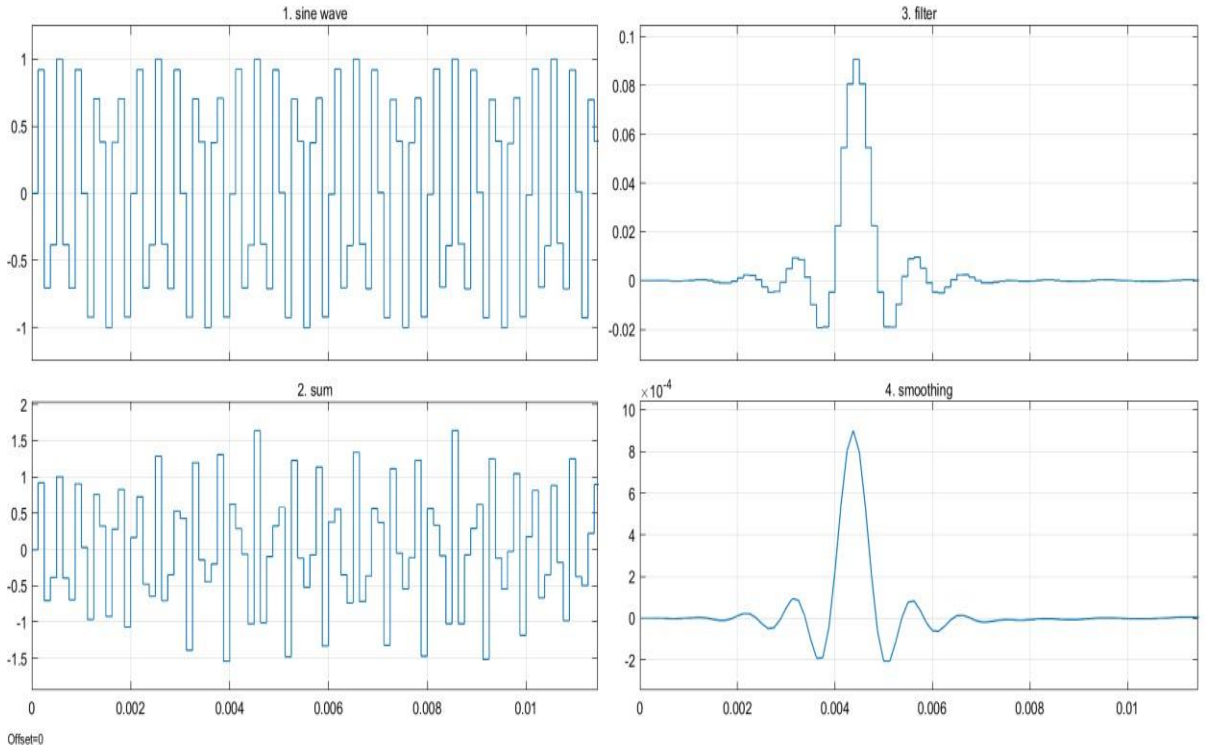
Passband – 0.5kHz



Transition – 1.2kHz

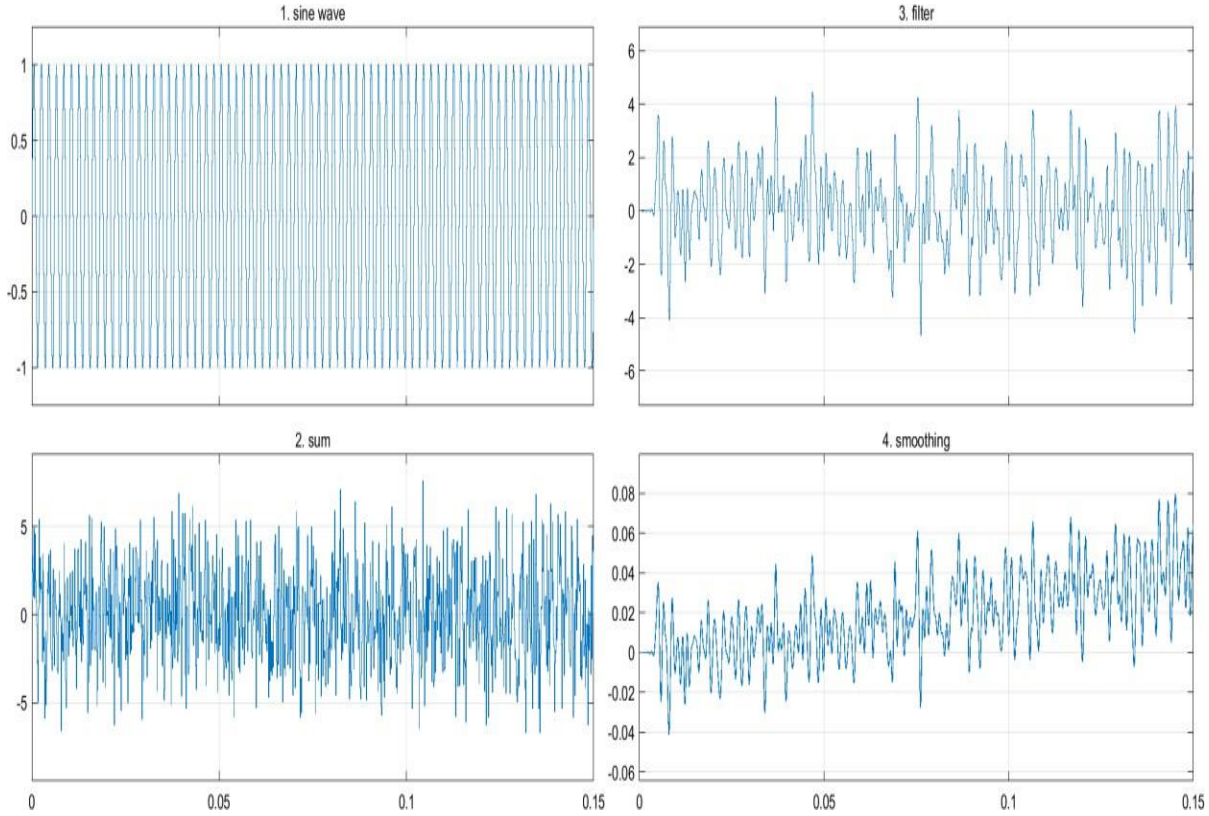


Stopband – 2.5kHz

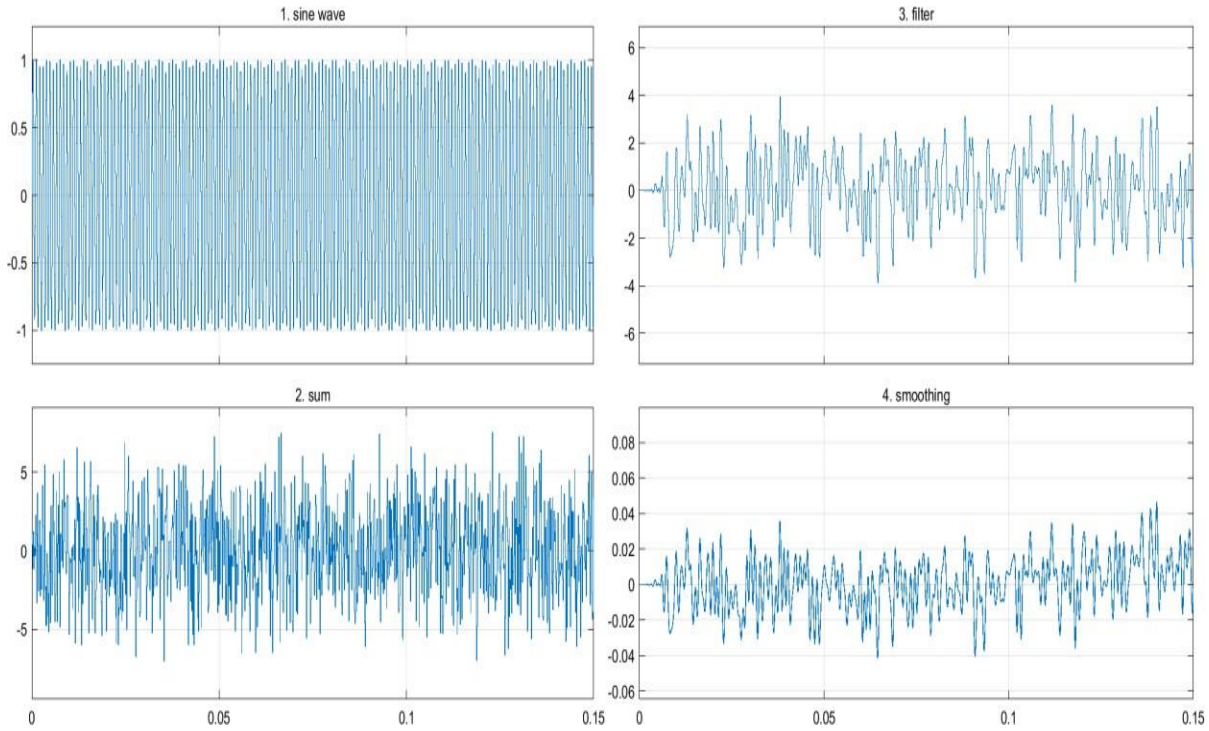


LOWPASS FILTER --FIR

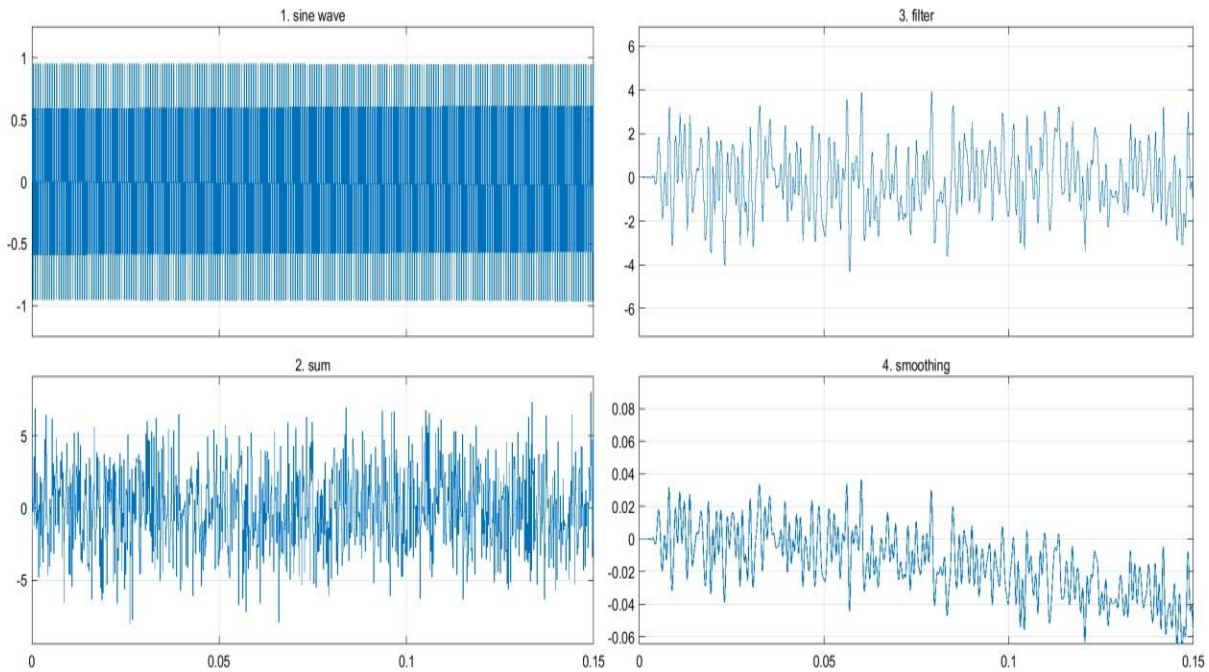
Passband – 0.5kHz



Transition – 1.1kHz

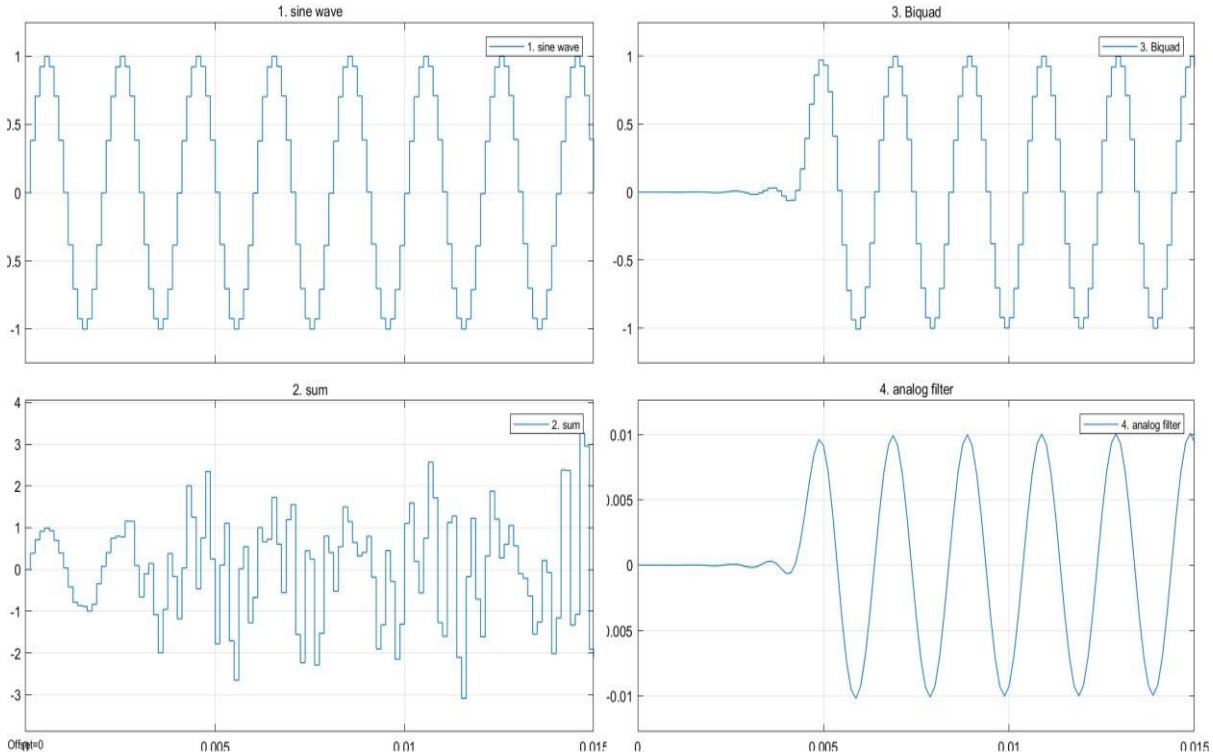


Stopband –3.2kHz

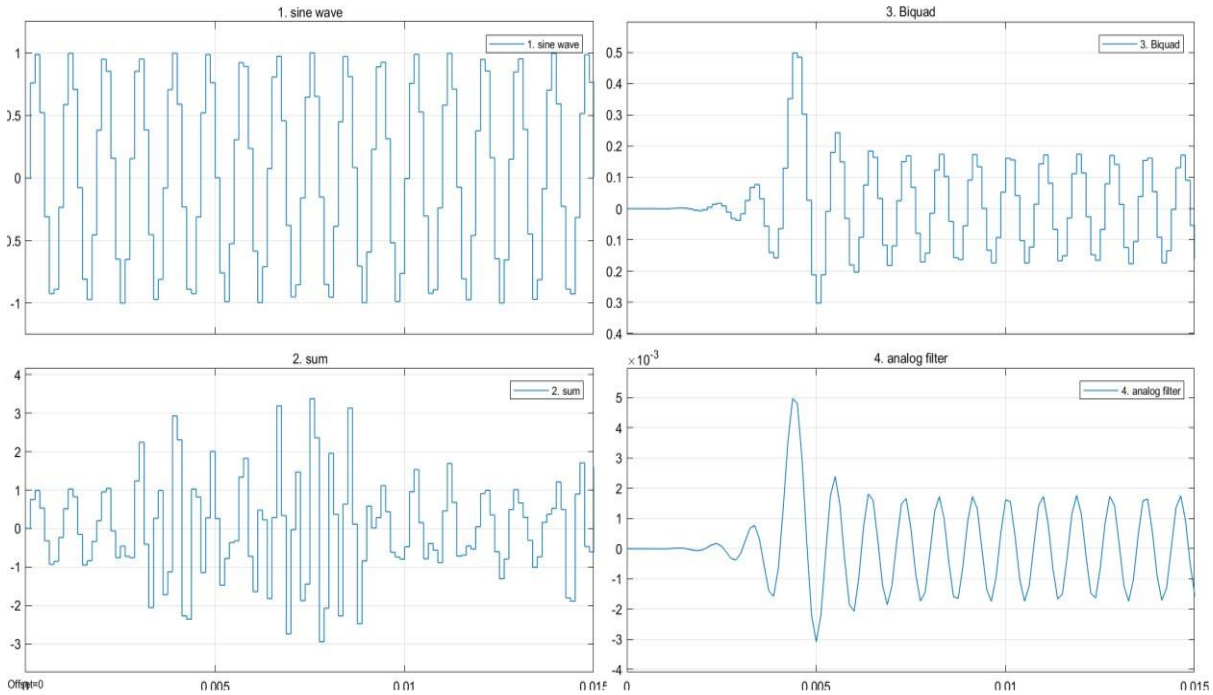


BANDPASS FILTER --FIR

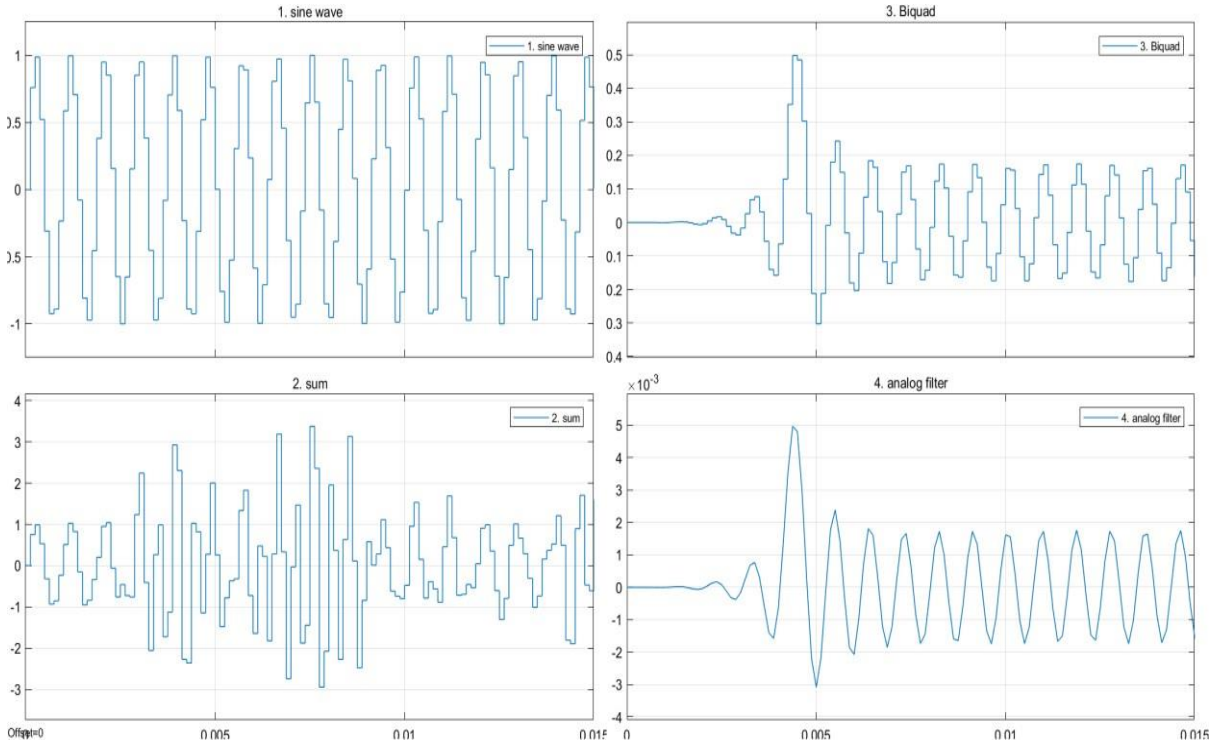
Passband – 0.5



Transition – 1.1kHz



Stopband – 3.5kHz


1. sine wave
1. sine wave
3. Biquad
3. Biquad
2. sum
2. sum
4. analog filter
4. analog filter
×10^-3

Discussion Questions

1. What is the order of the lowpass Butterworth lter you designed?

The order of the lowpass Butterworth filter designed in this lab is **26**. This order was chosen to achieve the desired frequency response while maintaining a balance between filter complexity and performance.

2. What is the order of the lowpass Chebyshev Type I lter you designed?

The order of the lowpass Chebyshev Type I filter designed in this lab is **12**. Chebyshev filters typically achieve a sharper roll-off compared to Butterworth filters with a lower order, which is reflected in the design.

3. Compare the memory usage of each of the 3 lters assuming a Direct Form II realization is employed. Do you see how ine cient the windowing technique is? How much more expensive in terms of memory is the windowing technique from the best IIR lter?

In a Direct Form II realization, memory usage is indicated by the number of adders required:

- **Butterworth IIR filter**: 52 adders.
- **Chebyshev IIR filter**: 24 adders.
- **FIR filter (Blackman window)**: 58 adders.

The windowing technique, specifically the Blackman window, uses the most memory with 58 adders. This highlights its inefficiency, as it requires more memory compared to both the Butterworth and Chebyshev filters. The Blackman filter achieves the required performance only with a very high order, leading to increased memory usage. In comparison, the Chebyshev filter is the most efficient, requiring the fewest number of adders. Thus, the windowing technique (Blackman FIR filter) is significantly more expensive in terms of memory than the best IIR filter (Chebyshev), which makes it less favorable for memory-constrained applications.